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Corso di laurea in Informatica

Citation Attribution at Claim Granularity in RAG Systems: Comparing NLI and LLM Approaches

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Anno Accademico 2025–2026

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Abstract

Large Language Models (LLMs) generate fluent answers but routinely produce plausible, unfounded statements. Retrieval-Augmented Generation (RAG) grounds answers in retrieved passages, yet grounding helps the reader only when it is visible, that is, when each statement carries a citation to the passage that supports it. This thesis addresses *citation attribution*: given an LLM-generated answer and its context passages, identifying, for each atomic claim of the answer, the passages that support it. We formalize the task as an attribution function φ and, under oracle retrieval, compare two contrasting implementations spanning the trade-off a real deployment faces: a capable general-purpose LLM (*Claude Haiku 4.5*) against a lightweight Natural Language Inference (NLI) cross-encoder (*DeBERTa-v3-large*) that runs locally, at zero cost and reproducibly. The comparison runs on P-Cite, a modular pipeline (generation, decomposition, attribution, citation) with interchangeable attributors. Since existing benchmarks lack claim-level ground truth, we introduce ALCE+, a claim-level benchmark built on ALCE around hand-annotated atomic facts (*nuggets*), complemented by an independent LLM judge; the two evaluation lenses agree on weak configurations and diverge, interpretably, on strong ones, validating the framework itself. The comparison, run on three question regimes, yields a trade-off rather than a winner. The plain cross-encoder is severely distracted by injected retrieval noise: up to two thirds of its claims cite distractors, and its judge-assessed precision drops as low as 40%; the LLM is immune. Gating the cross-encoder’s candidate set with a cheap bi-encoder pre-filter removes the distraction almost entirely and makes the local model the most precise attributor in the comparison (92–99% judge-assessed precision against the LLM’s 89–94%) at nearly no cost in coverage, while the LLM retains higher recall throughout, by ten to twenty points, at roughly four cents per question against under one for the local model. An error analysis traces the ungated failures to a structural entailment bias of the DeBERTa model. The full implementation is released as an open-source web application.¹

¹<https://github.com/simoninoProgrammatore/post-generation-citations-reactversion>

Chapter 1

Introduction

1.1 LLMs and Retrieval-Augmented Generation

Large Language Models (LLMs) are neural networks trained on massive text corpora to predict the next token of a sequence; at sufficient scale, this simple objective yields systems capable of answering questions, summarizing documents and sustaining articulate conversations in natural language. Their answers are fluent and, at least in appearance, authoritative. That appearance, however, is no guarantee: LLMs routinely produce statements that are plausible yet unfounded, a phenomenon known as *hallucination* [6], and they do so with the same confident tone with which they report correct facts. To a non-expert reader, the two are indistinguishable.

The most widespread mitigation is the *Retrieval-Augmented Generation* (RAG) paradigm [7]. Instead of relying solely on the knowledge encoded in its parameters, the model is paired with a retrieval module that, given the user’s question, collects relevant passages from a document corpus and provides them as context for generation. The answer is thus, in principle, grounded in verifiable sources, though not infallibly so: the model can still assert claims absent from the retrieved passages, and even when the content is faithful it may attach citations that do not actually support it. Grounding, however, is useful to the reader only when it is *visible*: hence the growing interest in generating answers with *citations*, where every statement carries an explicit reference to the passage that supports it, following the principle of *Attributable to Identified Sources* [16]. The task of assigning to a piece of text the source that supports it is called *attribution*, and it is the task this thesis is about. Benchmarks such as ALCE [4] have made it measurable, defining the *citation precision* and *citation recall* metrics and elevating citation quality to an object of systematic evaluation.

1.2 The problem

Attribution, in the form this thesis studies it, is stated as follows: given an answer generated by an LLM and the set of passages provided as context, identify, for each information unit of the answer, the passages that support it. The following example illustrates the task.

Running example

Question. Who discovered penicillin?

Generated answer. Alexander Fleming discovered penicillin in 1928 at St Mary’s Hospital. He was a Scottish physician.

Coarse attribution (whole answer). The answer as a whole is traced to the documents it draws on, without saying which fact comes from where: → *Alexander Fleming, History of penicillin.*

Claim-level attribution. The answer is decomposed into atomic facts, and each is attributed on its own.

“Fleming discovered penicillin” → “... in 1928, Alexander Fleming discovered ... the substance he named penicillin...”

while a secondary fact such as “Fleming was Scottish” is supported by a different passage, “... A. Fleming, Scottish physician, born in Lochfield...”

The example shows the two readings of the same task. In general, attribution is the problem of establishing whether a statement is supported by an identified source. The granularity at which the statement is taken, from the whole answer down to the single atomic fact, is a design choice. Coarse attribution links the answer to its sources but cannot tell which fact each one supports; claim-level attribution recovers, for every fact, the passage that supports it, and distinguishes the central claim that answers the question from the incidental ones. This thesis works at the finest of these granularities, the atomic claim. This example is illustrative and hand-constructed for clarity; a complete, real trace produced by the system on an actual benchmark question, together with its full evaluation under both lenses, is reported in Appendix A.

The recent literature, systematized by the survey of Schreieder et al. [19], identifies several paradigms for the task of attribution; two of them are central to this work. In the *pre-hoc* (or *generation-time*) approach the model produces citations together with the answer, while in the *post-hoc* (or *post-generation*) approach the answer is generated freely and attribution takes place afterwards, as a separate step [18]. This thesis is situated in the second paradigm, which offers two structural advantages: it decouples the quality of

generation from that of attribution, making the two components analyzable in isolation, and it applies to any answer already produced, without requiring modifications to the generating model.

We already announced that we operate at the finest granularity proposed in the literature: that of the *atomic claim*. Following the approach of FActScore [13], the answer is decomposed into minimal, self-contained and independently verifiable facts, and it is to each of these facts, rather than to the answer as a whole or to the single sentence, that attribution assigns its evidence. The reason is one of precision: a single sentence often bundles several facts, and a citation attached to the whole sentence may be correct for one of them and wrong for another. Only by decomposing the answer into atomic facts can support be assessed, and credited or denied, one fact at a time, which is exactly what makes the attribution informative enough to be evaluated rigorously. Formally, the object of study is an attribution function φ that maps each claim of the answer to the subset of passages supporting it. The thesis adopts an *oracle retrieval* assumption (the relevant passages are given), so that retrieval errors do not confound the comparison; the answer is instead generated by the pipeline itself, with the generation step held identical across the attribution methods under study.

This thesis studies two families of methods for implementing φ , chosen because they embody the two natural and contrasting ways of deciding whether a passage supports a claim. The first uses *Natural Language Inference* (NLI) models, typically cross-encoders such as DeBERTa-v3 [5], which classify each (passage, claim) pair as entailment, contradiction or neutrality; NLI is the standard proxy for textual support, and is the criterion the ALCE benchmark itself adopts to judge citations [4]. Such models are small, encoder-only transformers, well below one billion parameters (DeBERTa-v3-large, the model adopted here, has 435 million), and can therefore be run on ordinary consumer hardware: they execute locally, at zero cost, deterministically and reproducibly, but operate at sentence granularity and are sensitive to surface form. The second uses flagship LLMs, reached through a commercial API, as attributors: they judge support with richer reasoning capabilities and extract the supporting evidence directly, but are costly, non-deterministic and dependent on an external provider. The two thus sit at opposite ends of a cost and capability trade-off, and comparing them, in terms of how wide the quality gap is and where the local method fails, is the research question around which this thesis is built.

1.3 Proposal and rationale

The objective of this thesis is to measure how far a cheap, local NLI attributor can go, relative to a more capable but costly LLM attributor, when both are asked to attribute individual atomic claims under identical conditions. Concretely, we ask: at the granularity of the atomic claim, and with the rest of the pipeline held fixed, how wide is the quality gap between the two methods, and where exactly does the local one fail? Framing the comparison around what is achievable with an efficient method also motivates the error analysis of the NLI attributor in Chapter 6: characterizing where the cheap method breaks is part of answering the question.

To run this comparison we design and implement *P-Cite*¹, a modular post-generation pipeline in four steps (generation, decomposition, attribution and citation), followed by a separate evaluation stage, exposed through a REST architecture with interchangeable components. The attribution step accepts either of the two methods under study, so that NLI and LLM attributors can be swapped while everything else in the pipeline is held fixed: any difference in the final citation quality is then imputable to the attribution function alone. Figure 1.1 gives an overview of the pipeline; Chapter 3 describes each step in detail.

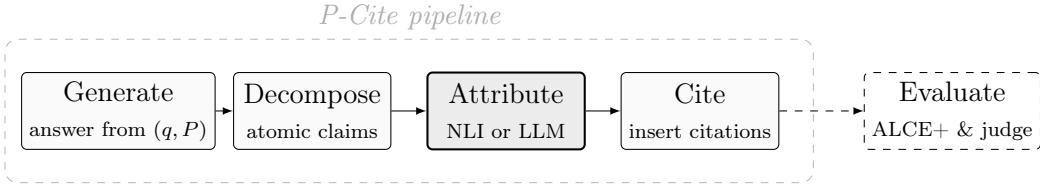


Figure 1.1: Overview of the P-Cite pipeline. The four stages Generate, Decompose, Attribute and Cite form the pipeline proper (dashed box); evaluation is a separate stage, added on top to measure the result, drawn detached with a dashed outline and a dashed handoff. The attribution step (highlighted) accepts either of the two methods under comparison; all other stages are identical across configurations.

The output of the system is the original answer annotated with inline citation markers and an interactive viewer that, for each sentence, exposes the underlying claims, the supporting passages and the extracted evidence

¹The name P-Cite is also used in [18] to denote the post-hoc citation paradigm in general, as opposed to generation-time citation (G-Cite). Our pipeline is an independent implementation of that paradigm, and we keep the name for consistency with this terminology.

highlighted in the text. The full implementation is released as an open-source web application.²

Comparing the two attributors, however, requires a reliable measuring instrument, and none of the existing benchmarks provides one in the form this task needs. KILT [14] annotates provenance at the page level, too coarse for the atomic claim. AttributionBench [9] fixes the claim–evidence pairs, making them unusable for claims generated by a different pipeline. ALCE [4] evaluates citations through an NLI judge, which raises a problem of *circularity*: since one of the attribution methods we want to evaluate is itself NLI-based, the system would be measured against itself, and the judge’s systematic errors would become invisible. Also, ALCE does not have the granularity that we use for our attribution function, in fact, it lacks claim-level ground truth.

This gap motivates the evaluation we build for the thesis. Inspired by ALCE, we assemble *ALCE+*, a claim-level dataset whose core is a human gold standard: for a set of benchmark questions, the atomic facts that a correct answer should contain (*nuggets*) are annotated by hand. Since the metrics that we will define score only the claims that match an annotated nugget, we add a second, complementary lens with total coverage: an LLM judge that assesses the logical support of every claim–evidence pair. The choice rests on the LLM-as-a-judge methodology, whose judgments have been shown to agree with human ones at a rate comparable to the agreement between human annotators themselves [25]. The two lenses answer different questions, fidelity to a human gold standard on one side and logical support on the other, and together they make the NLI vs LLM comparison robust to the weaknesses of either.

1.4 Contributions

The thesis offers three contributions.

P-Cite, a modular pipeline, and the NLI vs LLM comparison. We implement the four-step post-generation pipeline described above and use it as the experimental apparatus for our first empirical contribution: a systematic comparison of the two families of attributors, at atomic-claim granularity, under both evaluation lenses.

²<https://github.com/simoninoProgrammatore/post-generation-citations-reactversion>

ALCE+, a claim-level evaluation framework. We provide the hand-annotated nugget ground truth and the metrics computed against it, complemented by the LLM-as-judge measure: a judge-independent instrument for evaluating attribution functions at the granularity of the atomic fact.

An empirical analysis of the NLI attributor’s bias. Since the question we set out to answer is what an efficient, local method can achieve, understanding where that method breaks down is part of the answer. While working on NLI-based attribution, we observed that the DeBERTa-v3 cross-encoder produces high-confidence entailment on certain premise–hypothesis pairs lacking any logical connection. The evaluation metrics make the resulting failures directly measurable, and an error analysis traces a substantial part of the NLI attributor’s mistakes back to this structural bias, which we characterize through documented cases, systematic ablation tests and mechanistic interpretability experiments (attention flow analysis, Integrated Gradients [20], activation patching [12]).

1.5 Thesis outline

The remainder of the thesis is organized as follows. Chapter 2 introduces the attribution problem, its granularities and the paradigms in the literature, and presents the two attribution methods under comparison together with the related work. Chapter 3 states the attribution task formally and describes the P-Cite pipeline step by step. Chapter 4 examines existing benchmarks and their limitations and introduces ALCE+, the claim-level benchmark built on ALCE: the nugget as the unit of truth, the required and optional distinction, the injected distractors, and how the gold standard was constructed. Chapter 5 defines how citation quality is measured on top of that gold: the two-stage claim-to-nugget matching, the two evaluation lenses (the nugget lens and the complementary LLM-as-judge lens), and the metrics each of them reports. Chapter 6 reports the experimental setup and the results of the NLI vs LLM comparison, including the effect of noise injection and the error analysis that uncovers the structural bias of the NLI attributor. Chapter 7 summarizes the contributions, discusses their limitations and outlines future directions.

Two appendices complete the thesis. Appendix A reports a complete run of the pipeline on a single real benchmark question, from the generated answer through decomposition and attribution to the citations it produces and their evaluation under both lenses, so that every stage can be inspected on genuine system output. Appendix B collects the mechanistic interpretability

analysis behind the error analysis of Chapter 6, tracing the entailment bias of the DeBERTa cross-encoder through its internal computation.

Chapter 2

Background and Related Work

Attribution matters because the answers that motivate it cannot be taken on trust. Large Language Models produce fluent text that is frequently unsupported, and even when Retrieval-Augmented Generation grounds an answer in real sources, that grounding helps a reader only if each statement can be traced to the passage that supports it. Deciding which source supports which part of an answer is the attribution problem, and the techniques for posing and solving it are the subject of this chapter.

This chapter provides the conceptual and technical background of the thesis. Section 2.1 defines the attribution problem and the granularities at which it can be posed. Section 2.2 surveys the paradigms proposed in the literature for generating evidence-based text and positions our work among them. Section 2.3 introduces the model families that the thesis employs to assess support: NLI cross-encoders, sentence embedding models and LLMs, including their use as judges.

2.1 The attribution problem and its granularities

When an LLM produces an answer grounded in a set of sources, the *attribution* problem asks which source supports which part of the answer. The foundational formalization of this notion is the *Attributable to Identified Sources* (AIS) framework of Rashkin et al. [16]: a statement s is attributable to a source S if a generic reader would agree with the claim “according to S , s ”. AIS turns an intuitive idea of textual support into an operational test that human annotators can apply consistently, and it underlies most subsequent work on citation quality, including the metrics of the ALCE benchmark [4].

The same problem can be posed at three granularities, in increasing order of fineness.

1. **Answer to document.** The coarsest level: the answer as a whole is attributed to the document that justifies it, without distinguishing which fact comes from where. This is the level of *provenance* in knowledge-intensive benchmarks such as KILT [14].
2. **Sentence to sentence.** The intermediate level, adopted by most RAG systems with citations: each sentence of the answer carries one or more markers $[n]$ pointing to the evidences that support it. This is the granularity of ALCE [4].
3. **Atomic claim to sentence.** The finest level: the answer is decomposed into minimal information units, each attributed and evaluated independently against specific spans of evidence.

Running example: the three granularities

Answer to document. “Fleming discovered penicillin in 1928. He was Scottish.” → Wikipedia, *Alexander Fleming*.

Sentence to sentence. “Fleming discovered penicillin in 1928. He was Scottish.” [3] → [3] = “In 1928, at St Mary’s Hospital, Fleming...”

Atomic claim to sentence. The claim that answers the question, “Fleming discovered penicillin” → “...in 1928, Alexander Fleming discovered...penicillin...”, together with a secondary one, “Fleming was Scottish” → “...A. Fleming, born in Lochfield, Scotland, in 1881...”

This thesis operates at the third level. The finer the granularity, the more informative the attribution: a sentence-level citation may be correct for one fact in the sentence and wrong for another, and only claim-level attribution can tell the two apart. The price is that the answer must first be decomposed into claims, which raises the question of what exactly a claim is.

The atomic claim. What constitutes an atomic unit of information is a genuinely open question, with roots in the philosophy of language; no universally accepted definition exists. The most influential operational definition is that of FActScore [13], which evaluates the factual precision of long-form text by decomposing it into *atomic facts*: short, self-contained statements, each conveying a single piece of information and verifiable independently of the others. We adopt a working definition in the same spirit: an atomic claim is

any information unit whose corresponding question is either a yes/no question or a fill-the-blank question. For example, “Fleming discovered penicillin” is atomic, since it answers the single fill-the-blank question “Who discovered penicillin?” (or, equivalently, the yes/no question “Did Fleming discover penicillin?”), whereas “Fleming, a Scottish physician, discovered penicillin in 1928” is not, as verifying it takes several such questions. Operationally, the decomposition must also be *self-contained*: pronouns are resolved, the subject is always explicit, and references to the discourse context (“the passage”, “this person”) are forbidden, so that each claim can be assessed in isolation.

2.2 Paradigms for evidence-based generation

The literature on making LLM outputs attributable has grown rapidly, and the survey of Schreieder et al. [19], which analyzes 134 works, organizes it into seven approaches grouped in two families (Figure 2.1).

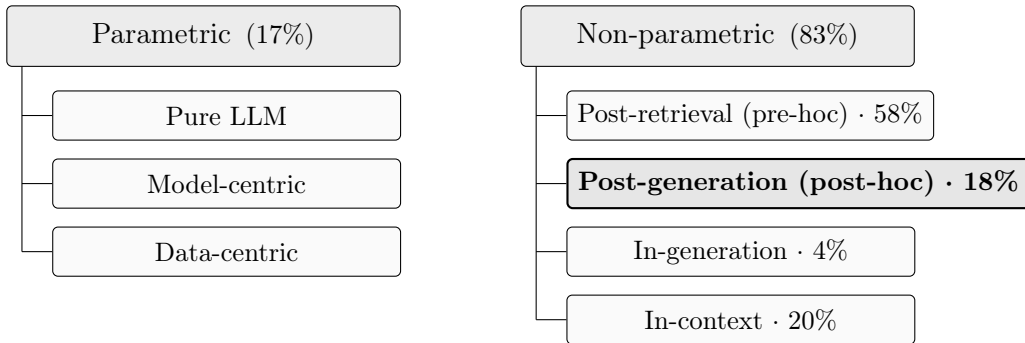


Figure 2.1: The taxonomy of attribution approaches of Schreieder et al. [19], with the share of the 134 surveyed works covered by each family and, for the non-parametric family, by each approach. This thesis is situated in the post-generation paradigm (highlighted).

Parametric approaches (around 17% of the surveyed works) rely on the knowledge encoded in the model parameters. They divide into *pure LLM* methods, which exploit the intrinsic abilities of an existing model without modification; *model-centric* methods, which alter the architecture or the training objectives; and *data-centric* methods, which curate or synthesize training data to improve attribution.

Non-parametric approaches (around 83%) integrate external evidence at inference time, and differ in *when* the evidence enters the process:

- *post-retrieval (pre-hoc)*, the standard RAG recipe and the most common approach (58%): evidence is retrieved before generation and passed to the model, which produces answer and citations together;
- *post-generation (post-hoc)* (18%): the answer is generated first, and evidence is retrieved and attached to each claim afterwards;
- *in-generation* (4%): the model decides if and when to retrieve during decoding;
- *in-context* (20%): the evidence is supplied directly in the prompt, with no retrieval step.

This thesis is situated in the post-generation paradigm. Its two structural advantages were anticipated in Chapter 1: attribution is decoupled from generation, so the two can be analyzed and improved in isolation, and the method applies to any answer already produced, including answers from models that cannot be modified or re-prompted. Its main limitation is symmetrical: a post-hoc attributor cannot prevent the model from generating unsupported content; it can only reveal it afterwards (and, optionally, remove it). The same decoupling, however, is also an opportunity: because the evidence is gathered independently of generation, the citations attached afterwards can be checked for consistency against the sources and, where they do not hold, corrected or improved before the answer reaches the reader.

Closely related work. Two lines of work are particularly close to ours. Citation-Enhanced Generation [8] is a representative post-hoc system: it retrieves evidence for each generated statement and verifies support with an NLI model, regenerating the response when claims are unsupported. Saxena et al. [18] compare the two paradigms directly, evaluating generation-time citation (G-Cite) against post-hoc citation (P-Cite) across four attribution datasets and finding a consistent trade-off between citation coverage and correctness. Our work differs from both in its object of study: rather than proposing a new attribution system or comparing paradigms, we compare *attribution functions* (NLI versus LLM) within the post-hoc paradigm, holding the rest of the pipeline fixed, and we focus on how to evaluate such functions reliably. To our knowledge this specific comparison, between attribution *functions* at atomic-claim granularity and with the pipeline held fixed, has not been carried out before; the pipeline itself follows the post-hoc recipe of systems such as CEG [8]. Vladika et al. [22] share our attention to the subtasks of answer segmentation and evidence retrieval, showing that more

self-contained claims improve attribution, which supports the decomposition constraints described above.

2.3 Transformer-based architectures

Every model used in this thesis, without exception, is built on the Transformer architecture [21]. This section is dedicated to those architectures and to the way the thesis uses them. It presents the three Transformer variants the work relies on, the decoder LLM, the NLI cross-encoder and the sentence-embedding bi-encoder, and, for each, the role it plays. The same architecture recurs in several guises and several roles, from generating the answer to deciding whether a passage supports a claim, from pre-filtering candidate evidence to judging the final output, and keeping those roles separate is itself part of the design.

Table 2.1 maps the architectures to their roles before the details. The role at the centre of the thesis, attribution, is filled by two of them, the NLI cross-encoder and the decoder LLM, and comparing these two attributors with the rest of the pipeline held fixed is the core question of the work. The LLM judge stands apart: it belongs to the evaluation of Chapter 5, not to attribution, and appears in this section only because it is the same architecture in a different role.

Architecture	Role in the thesis	Used in
Decoder LLM	Answer generator (RAG)	Ch. 3
Decoder LLM	Candidate attributor	Ch. 3
NLI cross-encoder	Candidate attributor	Ch. 3
Bi-encoder	Evidence pre-filter; semantic scoring	Ch. 3, 5
Decoder LLM	Evaluation judge (separate model)	Ch. 5

Table 2.1: The Transformer-based architectures used in the thesis and the role each one plays. Attribution, the focus of the work, is approached with two of them, an NLI cross-encoder and a decoder LLM, which the thesis compares with the pipeline held fixed.

2.3.1 Transformers and Large Language Models

The Transformer [21] is the common base of all three architectures. Its central mechanism, *self-attention*, lets every token of the input attend to every other token, building contextual representations in which the meaning of a word depends on its entire context. Large Language Models are Transformer

decoders trained on massive corpora with the next-token prediction objective; at sufficient scale they exhibit *in-context learning*: the ability to perform a task described or exemplified in the prompt, without any parameter update [3]. This makes them programmable through prompting, including *few-shot* prompting, where a handful of input/output examples in the prompt specify the expected behavior.

In this thesis the decoder LLM appears in three distinct roles. As the *generator*, it produces the answer from the question and the passages, following the RAG paradigm [7]. As a candidate *attributor*, it decides whether a passage supports a given claim, and in this role it is one of the two methods the thesis compares. As a *judge*, it evaluates the final output (Section 2.3.4). Keeping these roles separate is essential to the design: the judge is a different model from the generator and the attributor, which prevents a model from evaluating its own output.

2.3.2 Natural Language Inference and cross-encoders

Natural Language Inference (NLI) is the task of deciding, given a *premise* and a *hypothesis*, whether the premise *entails* the hypothesis, *contradicts* it, or is *neutral* with respect to it. Large annotated corpora such as SNLI [2] and MultiNLI [23] made it possible to train supervised models for the task, and NLI has since become the standard proxy for textual support: a passage supports a claim if it entails it. This is precisely how ALCE evaluates citation quality [4], and it is the principle behind the NLI attributor, the second of the two candidate attributors the thesis compares.

The strongest NLI models are *cross-encoders*: the premise and the hypothesis are concatenated into a single input, and the Transformer processes them jointly, so that every token of one text can attend to every token of the other. This full cross-attention is what gives cross-encoders their accuracy on pair classification, at the cost of requiring one forward pass per pair. The model used in this thesis is DeBERTa-v3 [5], an encoder that improves on BERT-style architectures through disentangled attention (separate representations for content and position) and an improved pre-training objective, fine-tuned on SNLI and MultiNLI for the NLI task.

Two structural limitations of NLI models matter for attribution. First, they operate at the granularity of the sentence pair: a passage must be split into sentences, each scored independently against the claim, with the scores aggregated afterwards. Second, being supervised classifiers trained on crowd-sourced pairs, they are sensitive to the surface form of the input, a weakness that Chapter 6 examines in depth.

2.3.3 Bi-encoders and sentence embeddings

Where a cross-encoder processes a pair jointly, a *bi-encoder* encodes the two texts separately into fixed-size vectors (*embeddings*) and compares them with a similarity function, typically the cosine similarity of the normalized vectors (Figure 2.2): the cosine of the angle between the two vectors, equal to 1 when they point in the same direction and to 0 when they are orthogonal, so that a higher value means the two texts are closer in meaning. The Sentence-BERT line of work [17] showed how to train encoders whose embeddings capture sentence-level semantics, making similarity search over thousands of sentences a matter of matrix operations rather than thousands of forward passes.

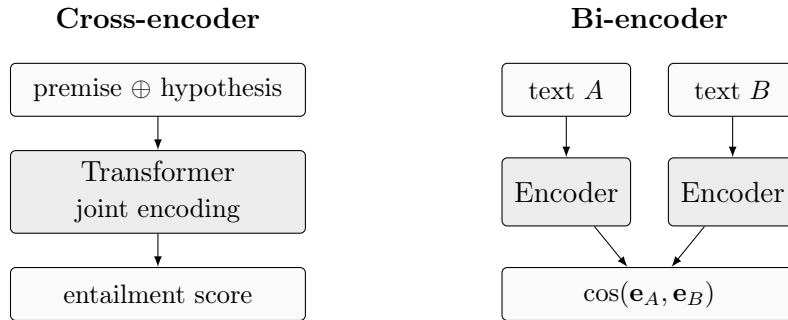


Figure 2.2: Cross-encoder versus bi-encoder. The cross-encoder processes the pair jointly with full cross-attention, yielding higher accuracy at one forward pass per pair; the bi-encoder encodes the texts independently and compares the embeddings, enabling cheap similarity search over large candidate pools.

The trade-off with cross-encoders is accuracy against scalability: a bi-encoder cannot model fine-grained interactions between the two texts, but it can score a claim against an entire corpus of candidate sentences at negligible cost. The two architectures are therefore complementary, and the thesis uses them as such, in the two places listed in Table 2.1: a bi-encoder pre-filters the candidate evidence pool before the expensive pair-level judgment (the specific encoders are reported in Chapter 3), and sentence embeddings provide the semantic component of the matching and scoring functions of the evaluation framework (Chapter 5).

2.3.4 LLM-as-a-judge

A recent evaluation paradigm employs an LLM to assess the quality of model outputs directly, a setting known as *LLM-as-a-judge* [25]. Its appeal is flexi-

bility: the judge needs no task-specific training, can follow articulated rubrics expressed in natural language, and can motivate its verdicts. Agreement with human annotators is high on many tasks, comparable to the agreement between two human annotators [25], which has made LLM judges a common substitute for expensive human evaluation.

The paradigm also has documented weaknesses: position bias (preferring answers by their position in the prompt), verbosity bias (preferring longer answers) and self-enhancement bias (preferring outputs of the same model family) [25]. This thesis uses an LLM judge in its evaluation, in the role already singled out in Table 2.1; the design choices that guard against these biases, such as scoring one claim/evidence pair at a time and drawing the judge from a different model family than both the attributor and the generator, are described where they belong, in Chapter 5.

Chapter 3

The P-Cite Pipeline

This chapter first states the attribution task as a well-defined computational problem, then describes P-Cite, the pipeline that implements it.

3.1 Problem formulation

The task this thesis studies is defined as follows.

Input. The input consists of:

- q , a question in natural language;
- $P = \{p_1, \dots, p_K\}$, a finite set of K documentary passages;
- a , an answer generated by an LLM conditioned on (q, P) , containing no citations.

Task. Construct an attribution function

$$\varphi : C(a) \rightarrow 2^P, \quad \varphi(c_i) = \{p \in P : p \models c_i\}, \quad (3.1)$$

where $C(a) = \{c_1, \dots, c_n\}$ is the decomposition of a into atomic claims, and $p \models c_i$ denotes that the passage p logically supports the claim c_i (an entailment relation).

Support is assessed at the granularity of the individual sentence. Each passage p is segmented into its sentences $\text{sent}(p) = \{s_1, s_2, \dots\}$, and the pool of candidate evidence is $E = \bigcup_{p \in P} \text{sent}(p)$. A passage supports a claim if and only if at least one of its sentences does:

$$p \models c_i \iff \exists s \in \text{sent}(p) : s \models c_i, \quad (3.2)$$

and the witnessing sentences constitute the *evidence* $\hat{e} \in E$ associated with the claim. Citations thus point to passages, while the supporting evidence is localized at the sentence level.

Output. The annotated answer a' , obtained by inserting into a citation markers $[j]$ that index the passages in $\bigcup_i \varphi(c_i)$.

Assumptions. Throughout the thesis, P is given (*oracle retrieval*): the retrieval stage is assumed correct, so that its errors do not confound the comparison between attribution methods. The answer a , by contrast, is produced by the pipeline itself; since the generation step is identical across the attribution methods under comparison, any difference in citation quality remains imputable to φ , which is the primary object of evaluation. The evaluation framework of Chapter 5 additionally monitors the content coverage of the generated answer, keeping the quality of generation observable without letting it interfere with the comparison.

How to evaluate the quality of φ is itself a non-trivial problem: it requires a ground truth at the granularity of the atomic claim and a judge independent of the methods under evaluation. This problem, and the framework we propose to address it, are the subject of Chapter 5.

3.2 Pipeline overview

P-Cite is a post-generation pipeline that turns a question and a set of passages into an answer annotated with citations. The pipeline is a cascade of four stages, *generate*, *decompose*, *attribute* and *cite*, each with a well-defined input and output, followed by the evaluation stage that is the subject of Chapter 5. Figure 3.1 shows the cascade with the data that flows between the stages.

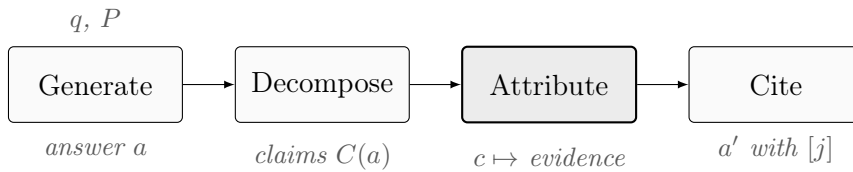


Figure 3.1: The four stages of P-Cite and the data exchanged between them. The attribution stage (highlighted) is realized by either the NLI or the LLM method; the other three stages are identical across configurations.

The guiding principle, stated in Chapter 1, is modularity: the stages are independent modules with JSON interfaces, so that the attribution stage,

which is the object of study, can be realized by either of two interchangeable methods while everything around it stays fixed. The implementation reflects this separation directly. The pipeline primitives live in a framework-agnostic core, exposed both as a command-line tool and as a REST service, so that a single stage can be run, inspected and cached in isolation. Table 3.1 collects the configuration of every stage; the rest of the chapter explains each in turn.

Stage	Component	Setting
Generate	model	Claude Haiku 4.5
Decompose	model	Claude Haiku 4.5
	passes	decompose + refine
Attribute	top- K per claim	3
NLI	cross-encoder	nli-deberta-v3-large
	pre-filter (sentences)	BAAI/bge-base-en-v1.5, hybrid
	pre-filter size	10 sentences
	support threshold τ	0.5
	aggregation	max over sentences
LLM	model	Claude Haiku 4.5
	decoding	greedy (temperature 0)
	pre-filter (passages)	all-MiniLM-L6-v2, hybrid
	pre-filter size	10 passages
	support	model verdict (no threshold)
	verdict	supports / contradicts / neutral
Cite	alignment	IDF-weighted containment

Table 3.1: Configuration of the pipeline stages used throughout the thesis. Cost-related figures are discussed in Chapter 7.

3.3 Generate

The first stage produces the answer that the rest of the pipeline will attribute. An LLM (Claude Haiku 4.5) is conditioned on the question and the K passages of the example, following the RAG recipe, and returns a plain-prose answer. Crucially, the prompt asks for an answer *without* citations: citation is a post-hoc step, and isolating generation from attribution is the entire point of the post-generation paradigm.

The prompt enforces three constraints that, although they may look like cosmetic formatting rules, each serve a downstream purpose.

- **No citations.** The model is told not to add markers such as [1] or [2]. Citations are inserted later, by the attribution stage, on the basis of

evidence rather than the model’s unverified self-report.

- **No meta-commentary.** Phrasings like “the passage says” or “according to the text” are forbidden, because they survive decomposition and produce degenerate claims such as “The passage says that Fleming was Scottish”, whose subject is the passage rather than Fleming.
- **Plain prose.** No markdown, headings or lists, so that the sentence segmentation used downstream operates on clean running text.

The prompt deliberately does not ask the model to reuse the wording of the passages. Such lexical grounding would push the generated claims to mirror the source text and could inflate both the surface sensitivity of the NLI method and the overlap-based metrics, masking the very weakness the thesis sets out to study; the answer is therefore generated in free prose.

The full prompt is the following.

PROMPT: GENERATION

Read the passages below and answer the question.

Passages:

[1] {title}: {text} ...

Question: {question}

Do NOT restate or echo the question, and do NOT add a title or heading. Begin straight with the answer. Answer the question directly and naturally, in plain prose. Start with what the user needs to know, then add a few additional facts from the passages that are relevant and useful. Do not write meta-commentary like ‘the passage says’ or ‘according to the text’. Do not add citation markers like [1] or [2]. Do not use any markdown: no headings (#), no bold, no bullet points, no lists. Write a single block of plain prose.

Answer:

Running example: generation

Question. Who discovered penicillin?

Passages (oracle). [1] *Alexander Fleming*; [2] *History of penicillin*; ...

Generated answer. “Alexander Fleming discovered penicillin in 1928 at St Mary’s Hospital in London. He was a Scottish physician.”

3.4 Decompose

The second stage breaks the answer into atomic claims, the units that the attribution stage will work on. The operative definition is the one given in Section 2.1: a claim is atomic if it is answerable by a single closed question, either yes/no or fill-in-the-blank. An LLM performs the decomposition under a few-shot prompt that encodes this definition as a set of rules, the most important of which are:

- **Atomicity.** One claim carries exactly one piece of information. If verifying a statement requires two questions, it is split into two claims.
- **Self-containment.** Every claim uses a real subject, never a pronoun: “He was Scottish” becomes “Fleming was Scottish”, so that the claim can be assessed in isolation.
- **No over-splitting.** Tightly bound coordinations and small closed lists are kept whole, to avoid fragments that lose meaning when separated.
- **Faithful wording.** Names, dates and numbers are kept verbatim, preserving the exact signals (a year, a figure) on which attribution often hinges.

A second *refine* pass re-examines each produced claim and splits any that is still not atomic, applying the same closed-question test. This two-pass design trades a second LLM call for a more reliable decomposition, which matters because every error here propagates to attribution: a non-atomic claim may be half-supported by one passage and half by another, with no correct attribution available.

The core of the decomposition prompt is the following; the two few-shot examples and the refine prompt, which re-applies the same closed-question test to each produced claim, are omitted for brevity.

PROMPT: DECOMPOSITION

Break the following text into atomic facts: short, self-contained statements that each carry exactly one piece of information.

RULES

1. ONE fact = ONE piece of information, answerable by a single closed question (yes/no or fill-in-the-blank). Ask yourself: "Can this be verified with one question?" If you need two questions, split it.
2. Use a REAL subject, never a pronoun. Never start with "It", "This", "They", "He", "She".

3. NEVER write meta-commentary. Do not start a fact with "The text", "The passage", "Historians propose that", or similar framing. State the fact directly.

4. Do NOT over-split. Keep a single fact whole when splitting would create fragments that lose meaning.

5. Keep specific names, dates, and numbers. Use the EXACT words of the source where possible.

6. Drop pure connective/filler sentences that carry no standalone fact.

Return ONLY a JSON array of strings, with no preamble or explanation.

Running example: decomposition

Input. "Alexander Fleming discovered penicillin in 1928 at St Mary's Hospital in London. He was a Scottish physician."

Atomic claims.

1. "Fleming discovered penicillin."
2. "Penicillin was discovered in 1928."
3. "The discovery of penicillin took place at St Mary's Hospital in London."
4. "Fleming was Scottish."
5. "Fleming was a physician."

3.5 Attribute

The third stage is the core of the thesis. Given the claims and the passages, it finds, for each claim, the evidence that supports it, realizing the attribution function φ of Section 3.1. Two methods implement φ and the pipeline treats them as interchangeable; the rest of the thesis compares them. As Figure 3.2 shows, the two methods share the same two-step shape, a cheap pre-filter followed by an expensive judge,¹ but they differ in what they pre-filter and in how they decide support. Both return, for each claim, the supporting passages with the evidence span that justifies each one.

¹The MiniLM pre-filter for the LLM comparison is a scalability mechanism for large candidate pools. Under the oracle retrieval of this thesis, each question carries about five passages, so the pool never exceeds the cut-off of ten passages and the pre-filter is effectively inactive: the LLM judge sees the full set, even when noise passages are injected. It would engage only in a true RAG setting that retrieves more than ten passages

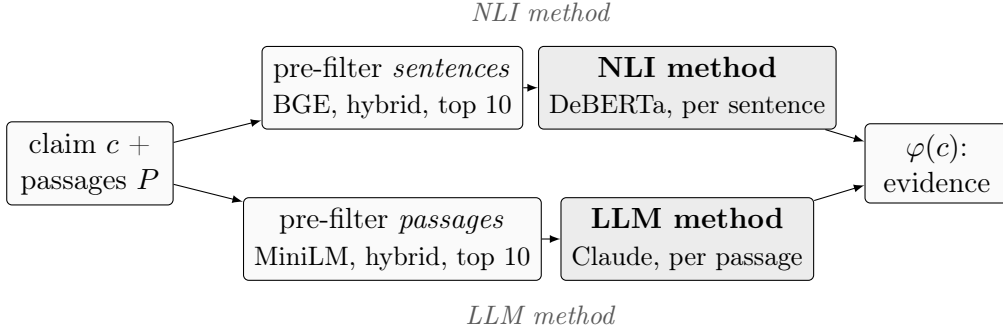
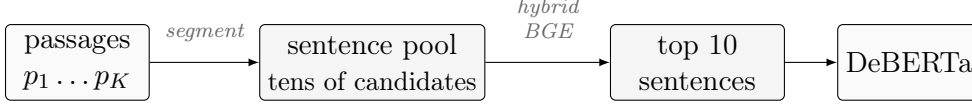


Figure 3.2: The two attribution methods share a pre-filter then judge shape but differ at both steps. The NLI method pre-filters *sentences* and judges each one with a cross-encoder; the LLM method pre-filters *passages* and judges each one with an LLM. Both pre-filters use the same hybrid score (Equation 3.3). Only one method is active in any given run.

The asymmetry of the two pre-filters deserves a justification (Figure 3.3). The unit each pre-filter ranks is dictated by the unit its judge consumes: the cross-encoder scores (sentence, claim) pairs, so the NLI path must select *sentences*, while the LLM receives whole passages in its prompt, so its path selects *passages*; pre-filtering sentences for the LLM would fragment exactly the context that gives it an advantage over the cross-encoder. The difficulty of the two rankings also differs. The sentence pool contains every sentence of every passage, tens of fine-grained candidates that are often close paraphrases of one another, and discriminating among them justifies the stronger encoder (BGE); the passage pool contains only the K passages of the example, a much coarser selection for which the lighter MiniLM is sufficient. Both rankings use the same hybrid score of Equation 3.3, so the two paths remain methodologically symmetric in *how* they rank and differ only in *what* they rank and with which encoder.

NLI method



LLM method

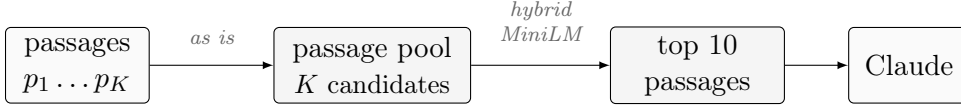


Figure 3.3: The two pre-filters compared. The NLI path segments the passages and ranks the resulting sentence pool; the LLM path ranks the passages themselves. Both use the hybrid score of Equation 3.3; the unit ranked matches the unit the downstream judge consumes.

3.5.1 The NLI method

The NLI method works at sentence granularity. Each passage is split into sentences by a rule-based segmenter, forming the candidate pool E of Section 3.1. The segmenter splits on sentence-final punctuation but adds two safeguards that keep the evidence spans exact: abbreviations are masked with same-length placeholders, so that character offsets computed on the masked text remain valid on the original, and fragments are re-merged when a candidate boundary falls before a digit or a lowercase letter, which protects decimals such as “8,848.86” and acronyms not in the abbreviation list from being split mid-sentence.

Scoring every sentence with the cross-encoder would be wasteful, so a bi-encoder (BAAI/bge-base-en-v1.5) first ranks the sentences against the claim and keeps the ten best for that claim. The ranking is a hybrid of two signals:

$$\text{prefilter}(c, s) = \frac{1}{2} \cos(\mathbf{e}_c, \mathbf{e}_s) + \frac{1}{2} \text{containment}(c, s), \quad (3.3)$$

where the first term is the cosine similarity of the sentence embeddings and the second is an IDF-weighted lexical containment, the fraction of the claim’s content tokens, weighted by inverse document frequency over the pool, that the sentence also contains:

$$\text{containment}(c, s) = \frac{\sum_{t \in T(c) \cap T(s)} \text{idf}(t)}{\sum_{t \in T(c)} \text{idf}(t)}, \quad (3.4)$$

with $T(\cdot)$ the set of content tokens of a text. The two signals are complementary: the embedding term captures paraphrase, where the words dif-

fer, while the containment term covers a known blind spot of embeddings, numbers, where nearly identical figures such as 8,848.86 and 8,849.96 yield nearly identical vectors although the exact digits are decisive. Being normalized on the claim, the containment is asymmetric and does not penalize long sentences, and rare pool tokens (dates, numbers, entities) dominate it; the tokenizer also normalizes numeric artifacts, so that “8,848. 86” matches “8,848.86”. The construction follows the containment queries of LSH Ensemble [26], computed exactly given the small pool. The pre-filter is tuned for recall: its job is not to discard a true supporter, while the decision of support is left to the judge.

The (sentence, claim) pairs are judged by the `nli-deberta-v3-large` cross-encoder of Section 2.3.2. For each pair the model produces logits over the three NLI classes; a softmax turns them into probabilities, and the entailment probability is the score of the pair. The support of a passage is the maximum entailment over its sentences, the continuous counterpart of the existential quantifier in Equation 3.2: a passage supports a claim if its best sentence does. Passages whose score clears the entailment threshold $\tau = 0.5$ are kept, and the top $K = 3$ of them, that is the three highest-scoring passages for the claim, are returned with the witnessing sentence as evidence. The threshold is set at the natural decision boundary of a probability, the point of majority mass, a precision-oriented choice consistent with the support criterion of ALCE [4]; its effect is examined with a sensitivity analysis in Chapter 6.

This method is local, deterministic and reproducible, and it runs at no API cost. Its weaknesses are equally structural: it works one sentence at a time, so support spread across two sentences is invisible to it, and being a supervised classifier it is sensitive to the surface form of the pair, a sensitivity that Chapter 6 shows can produce confident but spurious entailments.

3.5.2 The LLM method

The LLM method works at passage granularity. Here too a bi-encoder pre-filter (all-MiniLM-L6-v2) reduces the pool first, but it ranks *whole passages* rather than individual sentences, since the judge that follows reasons over a full passage at once. The ranking uses the same hybrid score of Equation 3.3, cosine similarity plus IDF-weighted containment, applied at passage granularity; the two attribution paths thus share the same pre-filter design and differ only in their unit (sentence versus passage) and in their judge.

The pre-filtered passages are then presented to an LLM (Claude Haiku 4.5) in a single prompt, and for each one the model returns a structured verdict, *supports*, *contradicts* or *neutral*, and, when the verdict is *supports*, the

exact sentence that constitutes the evidence. Unlike the NLI method, the LLM method has no numeric threshold: a passage is kept when the model labels it *supports*, and the evidence is the sentence the model returns. The supporting passages are kept, top- K as before, and a span-recovery step locates the returned evidence in the original passage so that it can be highlighted in the interface. The model is queried with greedy decoding (temperature 0), so its output is deterministic up to the residual non-determinism at the API level.

Compared to the NLI method, the LLM reasons over the whole passage rather than sentence by sentence, is robust to paraphrase, and extracts the evidence directly instead of relying on a separate scoring pass. The costs are the mirror image of the NLI method’s advantages: it depends on an external API, it is not free, and it is reproducible only up to the API-level non-determinism just noted. The magnitude of the cost is quantified in Chapter 7.

The prompt, including its single in-context example, is the following.

PROMPT: LLM ATTRIBUTION

```
You are a fact-checking assistant. Check if passages support a
claim.

Claim: "{claim}"

For each passage, decide: "supports", "contradicts", or "neutral".
If "supports", copy the EXACT sentence from the passage that supports
the claim.

EXAMPLE:
Claim: "The Eiffel Tower is 330 meters tall."
Passages:
[0] Eiffel Tower: The Eiffel Tower is a wrought-iron lattice tower
in Paris. It is 330 metres tall and was completed in 1889.
[1] Big Ben: Big Ben is the nickname for the Great Bell in London.
The tower is 96 metres tall.

Output:
[{"idx": 0, "label": "supports", "evidence": "It is 330 metres
tall and was completed in 1889."}, {"idx": 1, "label": "neutral",
"evidence": ""}]

Now analyze these passages. Return ONLY a JSON array, nothing else.

Passages:
{passages}
```

Running example: attribution

Claim. “Fleming discovered penicillin.”

NLI method. Passage [2], best sentence “...in 1928, Alexander Fleming discovered...the substance he named penicillin...”, entailment 0.95 $\Rightarrow$ supported.

LLM method. Passage [2], *supports*, evidence “in 1928, Alexander Fleming discovered... penicillin”.

Both methods attribute the claim to passage [2]; the chapters that follow measure how often, and how faithfully, they agree with a human ground truth.

3.6 Cite

The final stage takes the attributed claims and reinserts citation markers into the answer, producing the annotated answer a' together with an interactive viewer. The difficulty is one of alignment: claims are self-contained sentences produced by decomposition, and they must be matched back to the spans of the original answer they came from, so that the marker lands on the right sentence.

The alignment is purely lexical and deterministic. Each claim is matched to the answer sentences using the same IDF-weighted containment of Equation 3.4, this time between the claim and the sentences of the answer, with the IDF weights computed over the answer itself: rare tokens such as dates, numbers and entities dominate the score, which neutralizes the noise introduced by decontextualization (the resolved pronouns and added subjects of Section 3.4).² The assignment follows a cascade: a claim is attached to the answer sentence (or sentences, when the same information recurs) whose containment is highest, provided it is high enough; a window of two adjacent sentences is considered instead for claims that merge two statements, accepted only if it improves on the best single sentence by a clear margin; and below a floor score the claim is left unattached, since no marker is preferable to a marker on the wrong sentence. A guard prevents a claim from attaching to a sentence that supplies only its subject, the case where decontextualization moved an entity (“He was blind” becoming “Andrea Bocelli was blind”) into a sentence that carries no fact of its own. Since a claim can attach to more than one sentence and a sentence can carry several claims, the sentence-to-claim relation is many-to-many in both directions. Each cited passage is

²A claim is re-attached, and thus shown in the annotated answer, only if it has at least one supporting evidence; unsupported claims carry no citation and are not displayed.

then assigned a stable number, the markers are inserted after the sentences that carry the supported claims, and a reference list is appended.

The viewer renders the result for inspection: clicking a sentence reveals the claims it contains and, for each claim, not just *which* source supports it but *where*: the source passage is shown with the exact supporting sentence highlighted inside it, together with its entailment score, for the NLI method, or its support verdict, for the LLM method.

Running example: citation

Citation map. [1] *Alexander Fleming* · [2] *History of penicillin*.

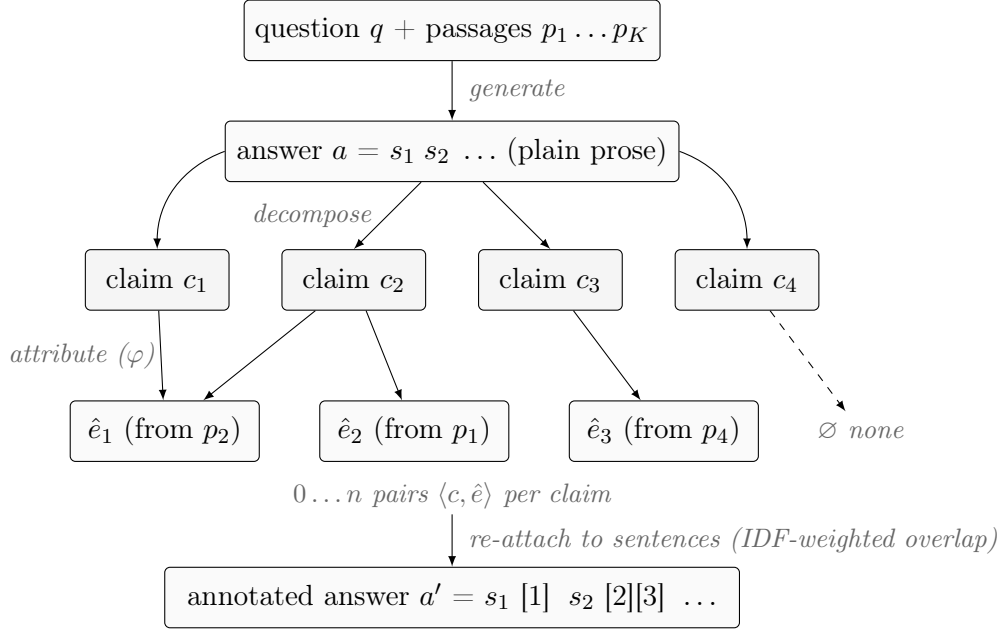
Annotated answer. “Alexander Fleming discovered penicillin in 1928 at St Mary’s Hospital in London [2]. He was a Scottish physician [1].”

Click-through. Clicking the first sentence shows the central claim “Fleming discovered penicillin”, its source [2] *History of penicillin*, and inside the passage the highlighted evidence “. . . in 1928, Alexander Fleming discovered. . . the substance he named penicillin. . .”.

3.7 Final overview

Having examined the stages one by one, it is worth retracing the data end to end (Figure 3.4). The pipeline received a question and its passages and produced a plain-prose answer (Section 3.3). That answer was decomposed into atomic claims (Section 3.4), and for each claim the attribution method in use, NLI or LLM, searched the passages for supporting evidence (Section 3.5): the result is, for each claim, between zero and n pairs $\langle c, \hat{e} \rangle$, where $\hat{e}$ is a sentence of some passage that supports c . A claim may be supported by several sentences, the same sentence may support several claims, and a claim may find no support at all.

The pairs, however, live at the level of claims, while citations must land on the sentences of the original answer. The cite stage (Section 3.6) therefore re-attached each claim to the answer sentence it came from, using the IDF-weighted token overlap of Equation 3.4; the claim carries its evidence along. This is what makes the interactive viewer possible: clicking a sentence of the annotated answer reveals the claims it contains and, for each claim, the source passage and the exact evidence span, highlighted in the original text.



click on a sentence $\rightarrow$ its claims, the source passage, the evidence span highlighted

Figure 3.4: The journey of an answer through P-Cite. The answer is decomposed into claims; attribution produces zero or more $\langle \text{claim}, \text{evidence} \rangle$ pairs per claim (with shared evidence possible, as for $\hat{e}_1$, and unsupported claims, as for c_4); claims are then re-attached to the answer sentences they came from, carrying their evidence, which enables the click-through inspection in the viewer.

With the answer annotated, the pipeline has done its work. What remains is to measure how well it did it: how accurate, and how faithful, the attribution produced by each method is. That measurement, and the framework built for it, are the subject of the next chapters.

Chapter 4

The ALCE+ Benchmark

The problem formulation of Chapter 3 left one question open: how to measure the quality of an attribution function φ . Doing so fairly requires two things that, as we will see, no existing benchmark provides together: a ground truth at the granularity of the atomic claim, and a judge that is independent of the methods under comparison. This chapter builds the first of the two. It summarizes the ALCE benchmark [4], the dataset our setting is built on, explains what it already provides and what it lacks, and then describes ALCE+, the enrichment we add on top of it to obtain a claim-level, fact-tied gold standard. The evaluation procedure that uses this gold, the nugget matching and the independent LLM judge, together with the metrics, is the subject of Chapter 5.

4.1 The ALCE benchmark

ALCE (Automatic LLMs’ Citation Evaluation) was the first benchmark to make answer generation *with citations* systematically measurable [4]. Its setting mirrors the one of this thesis: a model receives a question together with a set of retrieved passages and must produce an answer in which statements carry inline citations to the passages that support them.

Instance fields. An ALCE instance is a question together with a set of passages and a reference answer. Each passage carries a title, its text, and a retrieval score; the passages are the only evidence the model may cite, which makes the benchmark a closed-book attribution task over a fixed, given context. The reference answer, and, for the ambiguous-question dataset, a set of sub-questions with their gold short answers, are used to score correctness. For the question *Who is the original artist of “Sound of Silence”?*, for

example, the passages include the Simon & Garfunkel song, the homonymous 1966 album, and an unrelated Eurovision entry by Dami Im, and the reference answer disambiguates them.

The extraction field. Beyond title, text and score, each passage carries one field that is central to this thesis: an **extraction**. It can be filled or empty. When filled, it holds the verbatim span of the passage that is relevant to answering the question, that is, the exact sentence a human would point to as the evidence the passage contributes; when empty, it signals that the passage carries nothing useful for the question. The **extraction** field is therefore the closest thing ALCE offers to a ground truth for citations: it marks, passage by passage, which evidence supports the answer and in which words. Its limitation, which Section 4.2 develops, is that this evidence floats free of the facts it backs: the field says *this sentence is supporting evidence*, but never *this sentence supports that fact*.

Oracle retrieval. ALCE distinguishes a *retrieval* setting, in which the passages are the top results of a retriever, from an *oracle* setting, in which the passages are the relevant ones for the question. This thesis adopts the oracle setting: as stated in the problem formulation of Chapter 3, P is assumed given and correct, so that retrieval errors do not confound the comparison between attribution methods and any difference in citation quality is imputable to φ alone.

Question types. ALCE is assembled from three datasets that differ in the shape of a correct answer, summarized in Table 4.1. The three together stress attribution under different answer structures: a handful of short facts, a long list of items, and a multi-sentence explanation.

Dataset	Question type	Shape of a correct answer
ASQA	Ambiguous factoid	A few short answers, one per interpretation
QAMPARI	List	A long list of entity answers
ELI5	Long-form	A multi-sentence explanation (how / why)

Table 4.1: The three question types of ALCE, by the shape of a correct answer.

What ALCE measures. ALCE scores three dimensions: fluency, through MAUVE; correctness, through exact match of the gold short answers or recall of the gold sub-claims; and citation quality, which is the dimension of interest

here. Citation quality is split into a *citation recall*, whether the passages cited by a statement entail it, and a *citation precision*, whether each cited passage is actually needed. Crucially, both are computed by a Natural Language Inference model that decides whether a set of passages entails a statement, a design choice we return to in Chapter 5, where it turns out to be unsuitable for the comparison this thesis runs.

4.2 Why ALCE’s annotation is not enough

ALCE makes citation quality measurable, but the comparison this thesis sets out to run needs more than its gold annotation provides, and it needs a different judge than the one ALCE uses.

Evidence is not tied to facts. The `extraction` field does mark supporting evidence at the level of the passage, so it would be wrong to say ALCE offers no support annotation at all. What it does not offer is any link between that evidence and the individual facts of the answer. An extraction tells us that a passage contributes some relevant sentence, but not which atomic fact that sentence is meant to support, and there is no enumerated list of the facts a correct answer should contain. Two consequences follow. We cannot check whether the facts in a produced answer are the ones a correct answer was expected to state, because the expected facts are never spelled out. And we cannot verify that a produced citation backs the *right* fact, because the evidence in the gold is not addressed to any particular fact. Our attribution function works at the granularity of the atomic claim and returns, for each claim, an evidence span; ALCE, even with `extraction`, provides nothing at that granularity to match this output against.

The scoring is entangled with the methods. A second problem is not with ALCE’s annotation but with how it scores citations: the judgment is made by a Natural Language Inference model, and one of the two attributors we compare is itself NLI-based. Evaluating an NLI method with an NLI judge measures it against a close relative of itself. This is a problem about the evaluation procedure rather than the dataset, so we take it up, together with the independent judge that answers it, in Chapter 5.

The evidence-not-tied-to-facts gap is what the nugget enrichment of this chapter closes; the scoring gap is closed by the independent judge of Chapter 5.

4.3 ALCE+

ALCE+ reuses the questions, passages and reference answers of ALCE and enriches each instance at the level of the facts. It does not add a passage-level support annotation, because ALCE already provides one through `extraction`; what it adds, motivated by that field, is a layer of *nuggets*, the atomic facts a correct answer could contain, each tied to its own supporting evidence, plus an explicit treatment of *noise*. We illustrate the enrichment on the entry for the question *What caused World War I and when did it start?* It remains a dataset, not an evaluation procedure: the metrics of Chapter 5 are defined separately, on top of it.

4.3.1 From extraction to nuggets

The `extraction` field is the starting point, but only as a rationale, not as a source the nuggets are mechanically derived from. The reason is that the facts a correct answer must contain are defined by the question, and the questions in the benchmark are heterogeneous: what an extraction happens to lift from one passage is not, in general, the list of atomic facts the answer to that question requires. So we do not turn extractions into nuggets one for one. What we take from the extraction field is its idea, that there is a verbatim piece of evidence a correct answer should rest on, and we apply that idea at the right level: we enumerate the atomic facts a correct answer to the question must state, and we assign to each of them the citation it should receive, namely the exact golden evidence a system ought to extract for it. That assigned citation is what makes a comparison possible: it is the gold counterpart against which the evidence our own claims carry will be matched.

As a convenience, ALCE+ also restates, per passage, what an extraction already implies, in the explicit fields of Table 4.2: whether the passage supports the answer (`is_gold`), to what degree (`support_level`), through which exact sentence (`evidence_sentence`, the extraction span itself), and which sub-questions it answers (`supports_sub_questions`). These fields carry no information that the extraction did not; the genuine addition is the nuggets.

Field	Type	Meaning
<code>is_gold</code>	boolean	Whether the passage supports the answer
<code>support_level</code>	enum	Degree of support: full, partial, none
<code>evidence_sentence</code>	string	The supporting sentence (the extraction span)
<code>supports_sub_questions</code>	list	Indices of the sub-questions it answers

Table 4.2: Passage-level fields of an ALCE+ instance. They make explicit, in a queryable form, the support that the ALCE **extraction** field already signals; they are not a new annotation.

Figure 4.1 contrasts the two levels: on the left, the ALCE passage with its extraction, evidence not addressed to any fact; on the right, the ALCE+ nuggets, each an atomic fact with a golden evidence assigned as the citation it should receive. The arrow between them is a rationale, not a derivation.

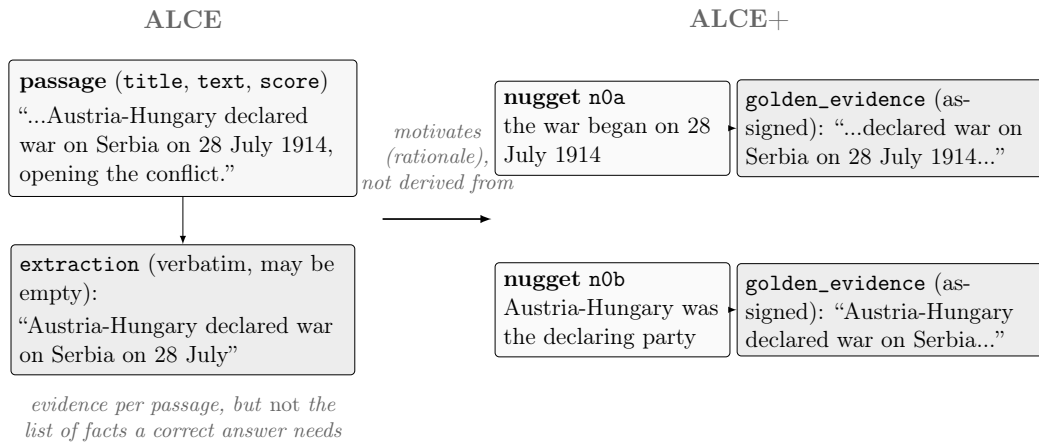


Figure 4.1: From the extraction rationale to fact-level evidence. The ALCE **extraction** (left) provides evidence per passage but not the list of facts a correct answer needs; that gap is the rationale for the ALCE+ nuggets (right), each an atomic fact with a golden evidence assigned as the citation it should receive. The nuggets are not derived one for one from the extraction, since the required facts are defined by the question, not by a single passage.

4.3.2 Nuggets: a claim-level gold standard

A *nugget* is an atomic fact that a correct answer to the question could contain. The notion follows the atomic-fact decomposition of FActScore [13], applied here not to a generated answer but to the *expected* content of a correct one. Each nugget carries the fields listed in Table 4.3: the fact itself, a set

of keywords, the title and index of the passage that supports it, the exact *golden evidence* sentence within that passage, and a flag marking whether the fact is required or optional.

Field	Type	Meaning
<code>text</code>	string	The atomic fact, in natural language
<code>keywords</code>	list	Salient terms of the fact
<code>golden_passage_idx</code>	integer	Index of the supporting passage
<code>golden_passage_title</code>	string	Title of the supporting passage
<code>golden_evidence</code>	string	The exact supporting sentence
<code>required</code>	boolean	Whether the fact is essential to a correct answer

Table 4.3: Fields of a nugget in ALCE+.

The golden evidence is the field that makes the gold standard usable for attribution: it anchors each expected fact to a specific sentence, so that the evidence a system produces can be compared against the evidence a human deemed correct, at the same granularity at which φ operates. How that comparison is carried out is defined in Chapter 5.

Required facts carry more weight. The `required` flag separates the facts a correct answer must contain from those that are relevant but accessory. This separation is not cosmetic: it lets the evaluation give more weight to the facts that are vital to the correctness of the answer than to marginal ones. Saying that World War I began in 1914 and was triggered by the assassination is part of what it *means* to answer the question correctly; mentioning the city of Sarajevo is a relevant detail but not a load-bearing one. Treating a system as well cited because it grounded an accessory fact, while missing a required one, would misrepresent it, so the evaluation reports coverage of the required nuggets separately and reads it as the primary signal (Chapter 5).

Atomic granularity in practice. The World War I entry makes the granularity concrete. Its reference answer opens with a single sentence, “World War I began on 28 July 1914, when Austria-Hungary declared war on Serbia”; ALCE+ splits it into two nuggets, one for the date (`n0a`) and one for the actor of the declaration (`n0b`), both anchored to the same golden evidence in the *World War I* passage. The assassination is likewise decomposed into separate facts: that it was the trigger (`n1a`), its date (`n1b`), its place (`n1c`) and its perpetrator (`n1d`). The split between required and optional then encodes what a correct answer must say versus what is merely relevant: the

date, the actor, the trigger and the perpetrator are required, whereas the city of Sarajevo (**n1c**) and the main long-term causes (**n2**) are optional.

Rationale. Nuggets answer both limitations of ALCE at once. They provide ground truth at the granularity of the atomic claim, tied to specific facts, which **extraction** alone does not; and, being a human annotation independent of any attribution method, they let the evaluation escape the circularity of an NLI judge, since the procedure that matches a produced evidence span against a golden evidence (Chapter 5) involves no NLI model at all.

4.3.3 Distractors and noise

Not every passage in an instance is gold. ALCE+ keeps, alongside the supporting passages, *distractors*: passages that are on-topic but do not support any required fact, with an empty extraction, marked by **is_gold** false and **support_level** none. In the World War I entry the biography of Franz Ferdinand is one such distractor: it is clearly about the right subject, yet it only repeats information already covered by the assassination passage and adds no new support. Distractors are deliberately hard, because a passage that is plainly off-topic would be trivial to reject; the interesting question is whether an attributor, faced with a passage that looks relevant but supports nothing, still cites only the genuinely supporting evidence or is drawn to the distractor. The noise metrics of Chapter 5 therefore report how many of the produced citation pairs draw their evidence from a non-gold passage, and how many of those still happen to match a golden evidence.

4.3.4 How ALCE+ was built

The dataset was built semi-automatically and then validated by hand. Starting from ALCE instances, and taking the **extraction** field as the rationale described in Section 4.3.1, the entries, that is the question, its passages and the set of nuggets with their golden evidence, were drafted by a large language model conditioned on the ALCE data. The automatic step produces a first draft of the nuggets; it is fast but not reliable on the dimension that matters most here, granularity. Every entry and every nugget was therefore inspected manually by the author, with two goals: that each nugget be genuinely atomic, neither bundling several facts nor splitting one across two, and that its golden evidence and its **required** flag be correct. The manual pass is what turns an LLM draft into a gold standard: the automatic generation provides coverage and scale, the human inspection provides the correctness and the granularity that the metrics rely on.

Chapter 5

Evaluating Citation Quality

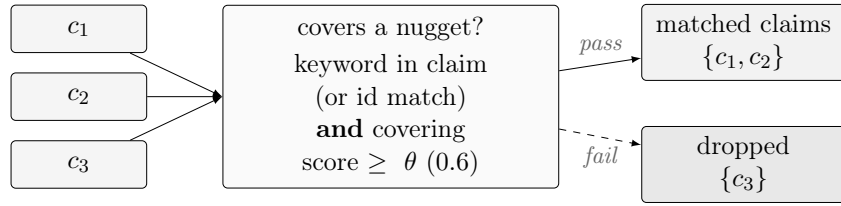
Chapter 4 built ALCE+, a gold standard at the granularity of the atomic claim. This chapter defines how the output of the pipeline is scored against it and, with it, how the two attribution functions of Chapter 3, the NLI cross-encoder and the LLM, are compared. We evaluate through two complementary lenses. The *nugget lens* matches the evidence a system produces against the human-annotated golden evidence of the nuggets; the *LLM-as-a-judge lens* asks an independent model whether the cited evidence supports the claim. The nugget lens needs no model and is anchored to the human gold, but it reaches a fact only through a matching step that can mispair a claim; the judge lens rules on every pair but replaces the human gold with a model’s verdict. Read together they are more informative than either alone. We first describe the nugget matching, then the judge, then see the two disagree on one example, and finally define the metrics.

5.1 The claim-to-nugget matching

The nugget lens compares the system output against the gold, and the comparison happens in two stages, shown in general form in Figure 5.1. The first stage is a *covering filter* on the pairing between a produced claim and a nugget. A claim covers a nugget only when two conditions hold together: a textual one, that at least one of the nugget’s keywords appears in the claim (or the nugget’s identifier is the one the pipeline attached to the claim), and a score one, that a covering score between the claim and the nugget reaches a threshold θ , set to 0.6. The claims that cover at least one nugget are the *matched claims*; a claim that covers no nugget, because it shares no keyword and clears no score, is set aside and is not scored against the gold. This stage answers *which of the produced claims are talking about a fact the gold*

standard knows about. The second stage looks only at the matched claims: for each, the evidence the system attached to the claim is compared against the golden evidence of the nugget it covers, through the similarity score σ of Section 5.4, and the citation counts as correct only when that comparison clears a second threshold τ . The two stages separate the two things a citation must get right, the fact and the evidence for it, and they are exactly the steps the metrics of Section 5.4 formalize.

Stage 1: covering filter (claim $\leftrightarrow$ nugget)



Stage 2: evidence check (matched claims *only*)

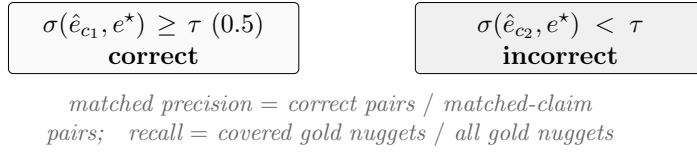


Figure 5.1: The two stages of nugget evaluation. Stage 1 is a covering filter on the claim-nugget pairing: a claim covers a nugget only if a keyword of the nugget appears in it (or the identifiers match) *and* a covering score reaches θ ; claims that cover no nugget are dropped. Stage 2 scores, for the matched claims only, the evidence the system attached against the golden evidence, counting a pair correct when the similarity σ clears τ . The matched precision is computed on the matched claims alone; the overall precision of Section 5.4 keeps the dropped claims in its denominator.

5.2 The LLM-as-a-judge lens

The nugget lens is anchored to a human gold and free of any model under test, but it reaches a fact only through the covering filter, and two things motivate a second, independent lens: the unsuitability of ALCE’s own judge, and a failure mode of the matching itself.

ALCE’s NLI judge is circular. ALCE judges citations with an NLI model: a citation is correct if the cited passage entails the statement. One of the two attributors we want to compare is itself NLI-based (Chapter 3). Evaluating an NLI attributor with an NLI judge measures the method against a close relative of itself: any systematic error shared by the two, in particular the entailment bias of the DeBERTa cross-encoder that we document in Appendix B, would be invisible to the evaluation, because the judge would make the same mistake as the method it is judging. A fair comparison needs a judge that coincides with neither method under test. The nugget lens already provides one, but it carries a failure mode of its own.

A weakness of gold matching. The covering filter is also the nugget lens’s main weakness, and it is the most concrete reason a second lens is needed. The pairing of a claim to a nugget is decided by shared keywords and a score, and it can go wrong: a claim may be paired, on a shared keyword, to a nugget that is actually about a different aspect of the topic. When that happens, the evidence the system attached, even if it is a perfectly correct citation for the claim, is then compared against the golden evidence of the wrong nugget, and the two will rightly look like they speak about different things, so the pair is marked incorrect. The mistake is not in the citation but in the gold pairing. A judge that reads the claim against its own cited evidence, and never against a nugget, is immune to this failure, which is exactly why we want it alongside the nugget lens rather than in its place.

Why a model as judge. Using an LLM as a judge of textual quality is by now a standard and reasonably reliable practice. On preference judgments, strong models agree with human annotators at rates comparable to the agreement between humans themselves [25]. The specific task here, deciding whether a span of text supports a statement, is exactly the kind of attribution judgment that recent work frames as the natural way to evaluate attributed generation [16], that automated attribution evaluators are increasingly used for [9], and that language models have been shown to perform competitively on [22]. The use of an automatic evaluator at the level of atomic facts also follows the precedent of FActScore [13], which scores fine-grained factual units automatically rather than by hand.

Independence. The decisive property of the judge, for our purposes, is independence. The judge model is neither of the two methods under comparison, so, unlike the NLI judge of ALCE, it does not share the entailment bias of the NLI attributor and does not measure that method against itself.

This is what lets the judge lens serve as a check on the nugget lens rather than a second view of the same bias.

Binary verdict. The judge reads each pair $(c, \hat{e})$ and returns one of two verdicts, `supported` or `not_supported`; a pair is correct in this lens when the verdict is `supported`. We deliberately avoid a graded or weighted verdict: a binary decision is the one an attribution claim ultimately reduces to, the cited evidence either backs the statement or it does not, and it keeps the lens simple to interpret and to compare against the nugget lens.

Cost. The two lenses also differ in cost, which is worth stating plainly. The nugget lens is a similarity computation against a fixed gold and is essentially free to run; the judge lens issues one model call per produced pair and is therefore the expensive part of the evaluation. This asymmetry is part of why the nugget lens, when it applies, is the primary instrument and the judge is the broader, costlier complement.

5.3 Two lenses on one example

Before the formal definitions, it helps to see the two lenses score the same output and, on purpose, disagree, since the disagreement is itself informative. Take three claims produced for the World War I question, each with one evidence span. Figure 5.2 shows the nugget lens and Figure 5.3 the judge lens, on the same three pairs.

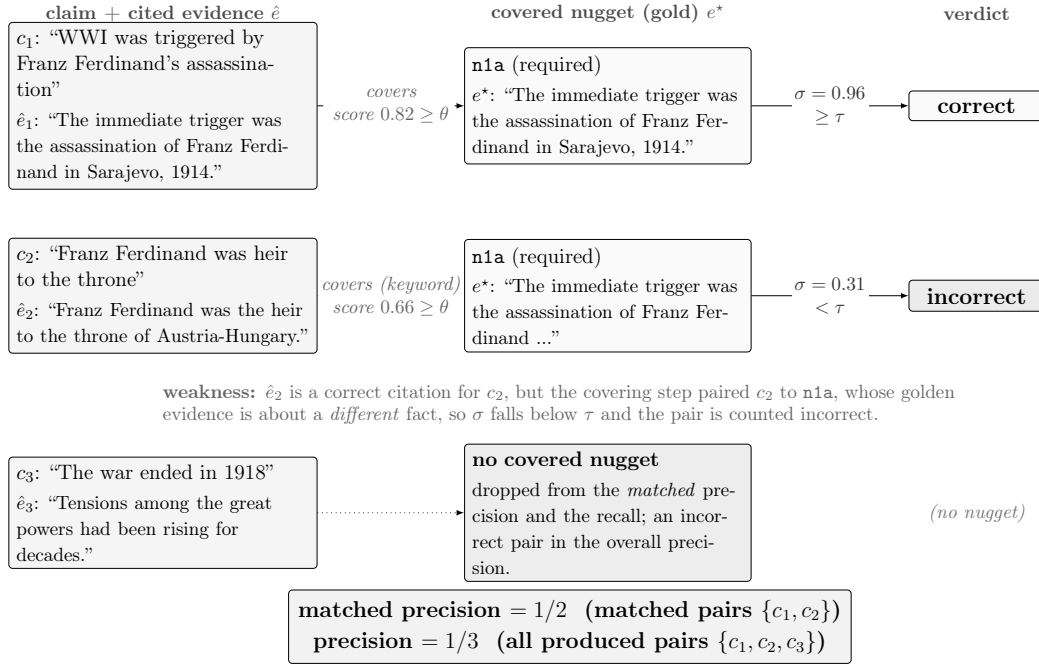


Figure 5.2: Nugget lens on the running example. Stage 1: c_1 and c_2 both cover **n1a** (a shared keyword and a covering score above θ), while c_3 covers no nugget. Stage 2, on the matched claims only: c_1 's evidence matches the golden evidence ($\sigma \geq \tau$, correct), c_2 's does not ($\sigma < \tau$, incorrect). One pair is correct, so the *matched* precision is $1/2$ over the two matched pairs, while the overall *precision* is $1/3$ over all three produced pairs, since c_3 's pair stays in the denominator although it covers no nugget. c_2 is the weak case: its citation is right, but the covering step paired it to a nugget about a different fact.

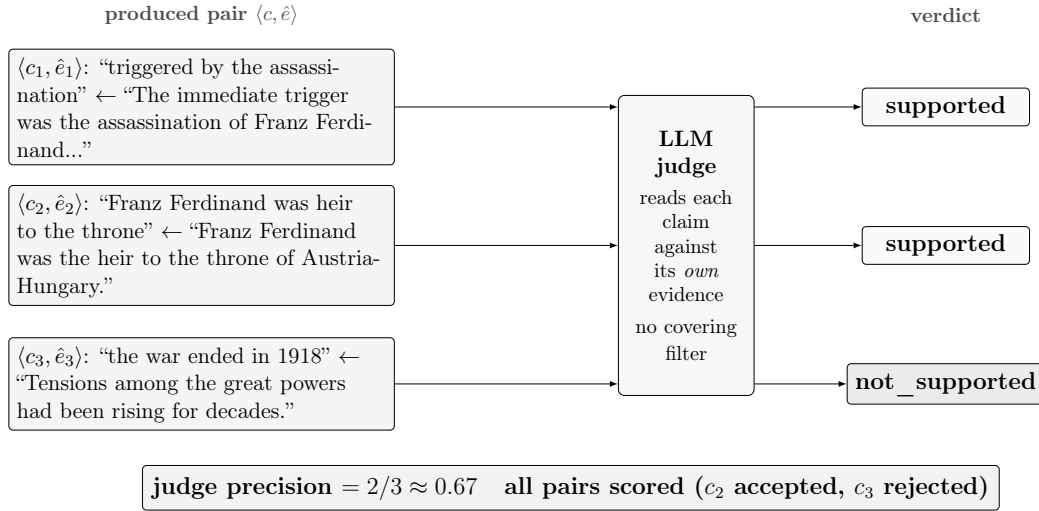


Figure 5.3: Judge lens on the same running example. The judge applies no covering filter and rules on every pair: c_1 and c_2 are supported, c_3 is not. It accepts c_2 , which the nugget lens rejected through a mismatched pairing, and it has a verdict for c_3 , which the nugget lens dropped, so it scores all three pairs.

The two lenses disagree by construction. The nugget lens, anchored to the annotated facts, scores the matched claims and reports a matched precision of $1/2$; counting all three produced pairs, its overall precision is $1/3$. The judge, over those same three pairs, reports $2/3$. The overall nugget precision and the judge precision share a denominator, so their gap, $1/3$ against $2/3$, isolates the single pair the lenses disagree on, c_2 : the judge accepts it, the nugget lens rejects it because the covering step paired it to a nugget about a different fact. On c_3 the two agree that the citation supports no fact. Neither number is the truth on its own: read together, they say where a method is faithful to the human gold and where the gold’s own pairing, rather than the method, is the limiting factor.

5.4 Evaluation metrics

We now define the metrics formally. The covering filter of Section 5.1 fixes which claims are in scope; on top of it the nugget lens reports two precisions and a recall, while the judge lens reports a single precision. The two lenses share one quantity, a precision computed over every produced pair, which makes them directly comparable; everything else is lens-specific.

Setup. Fix a question q . The pipeline produces a set of atomic claims $C(q) = \{c_1, \dots, c_n\}$ and, through the attribution and citation stages, attaches to each claim one or more evidence spans. Let $G(q)$ be the nuggets of q and $e^*(g)$ the golden evidence of nugget g .

Covering and matched claims. A claim c covers a nugget g when the textual and score conditions of Section 5.1 hold together, with the covering threshold θ set to 0.6:

$$\text{cover}(c, g) \iff [\text{kw}(g) \cap c \neq \emptyset \text{ or } \text{id}(g) = \text{id}(c)] \text{ and } s(c, g) \geq \theta, \quad (5.1)$$

where $\text{kw}(g)$ are the nugget’s keywords, $\text{kw}(g) \cap c$ are those that appear in the claim text, and $s(c, g)$ is the covering score the pipeline assigns to the pairing. The matched claims are those that cover at least one nugget,

$$M(q) = \{c \in C(q) : \exists g \in G(q), \text{ cover}(c, g)\}. \quad (5.2)$$

The unit and the two precisions. The unit of evaluation is the *citation pair* $(c, \hat{e})$, a claim together with one of its evidence spans. Two sets of pairs matter. The matched pairs are those of the matched claims,

$$\Pi(q) = \{(c, \hat{e}) : c \in M(q), \hat{e} \text{ a span attached to } c\}, \quad (5.3)$$

and the full set $\Pi_C(q)$ is defined the same way with $M(q)$ replaced by the full $C(q)$; since $M(q) \subseteq C(q)$, we have $\Pi(q) \subseteq \Pi_C(q)$. Precision over a set of pairs S is the fraction of correct pairs in it,

$$\text{precision}(S) = \frac{|\{(c, \hat{e}) \in S : (c, \hat{e}) \text{ correct}\}|}{|S|}, \quad (5.4)$$

where the predicate *correct* is fixed by the lens. The nugget lens reports this quantity over both sets: the *precision* $\text{precision}(\Pi_C(q))$, over every produced pair, and the *matched precision* $\text{precision}(\Pi(q))$, over the matched pairs alone. Since only a matched claim can produce a correct nugget pair, the two share a numerator and differ only in the denominator: a claim that covers no nugget contributes pairs to $\Pi_C(q)$ that can never be correct, so the overall precision is the stricter of the two and is lowered by off-topic citations, while the matched precision isolates the quality of the citations on the claims the gold knows about. The judge lens, which applies no covering filter, reports $\text{precision}(\Pi_C(q))$ with its own predicate (Section 5.4.2); computed over the same set, it is directly comparable to the nugget overall precision.

5.4.1 Nugget lens: fidelity to the human gold

In the nugget lens a pair $(c, \hat{e})$ is correct when its span matches the golden evidence of one of the nuggets that c covers. The match between a produced span $\hat{e}$ and a golden evidence e^* is a hybrid score

$$\sigma(\hat{e}, e^*) = \lambda \cdot \text{lex}(\hat{e}, e^*) + (1 - \lambda) \cdot \text{sem}(\hat{e}, e^*), \quad \lambda = 0.2, \quad (5.5)$$

where lex is an IDF-weighted token overlap and sem is the cosine similarity of sentence embeddings produced by a Sentence-BERT model (all-MiniLM-L6-v2). The lexical term rewards shared content words and exact figures, the semantic term tolerates paraphrase; the weighting leans on the semantic term while keeping the lexical one as a guard against superficially similar but factually different sentences. The pair is correct when this score clears the evidence threshold τ for some covered nugget:

$$(c, \hat{e}) \text{ correct} \iff \max_{g: \text{cover}(c, g)} \sigma(\hat{e}, e^*(g)) \geq \tau, \quad \tau = 0.5. \quad (5.6)$$

The maximum runs over the nuggets the claim covers, not over all nuggets: a claim is only ever scored against the facts it was matched to in Stage 1. Substituting this predicate into Equation 5.4 over $\Pi_C(q)$ and over $\Pi(q)$ gives the two nugget precisions; the recall is defined just below, over the gold.

Worked example. The running example of Figure 5.2 is exactly this computation. Of the three produced claims, c_1 and c_2 both cover **n1a** and c_3 covers no nugget, so $M(q) = \{c_1, c_2\}$ and $\Pi(q)$ holds two pairs while $\Pi_C(q)$ holds three. Only $e^*(\mathbf{n1a})$ against $\hat{e}_1$ clears τ ; $\hat{e}_2$ does not. There is therefore one correct pair, so the matched precision is $1/2$ over the two matched pairs and the overall precision is $1/3$ over all three produced pairs, c_3 's pair entering the denominator as an incorrect one. Recall, in the sense of Equation 5.7, instead looks at the gold: **n1a** counts as covered, since c_1 's correct pair matches its golden evidence, and it contributes to the numerator over the $|G(q)|$ nuggets at the denominator.

Recall. Precision measures what the system produced; recall measures how much of the gold it reached. Because the nuggets enumerate the facts a correct answer should contain, recall is naturally defined over them, not over the produced claims: it is the fraction of gold nuggets that some correct pair covers,

$$\text{recall}(q) = \frac{|\{g \in G(q) : \exists (c, \hat{e}) \in \Pi(q), \text{cover}(c, g) \text{ and } \sigma(\hat{e}, e^*(g)) \geq \tau\}|}{|G(q)|}. \quad (5.7)$$

This is a recall in the classical sense, with the gold at the denominator: it asks how many of the expected facts the system both stated and cited with matching evidence, and, unlike a denominator built from the system’s own output, it cannot be inflated by producing more citations. It is the quantity the implementation also calls the nugget coverage; the two are one and the same metric, not two. We report it overall and, more importantly, split over the required and the optional nuggets: as argued in Section 4.3.2, reaching the facts that are vital to a correct answer matters more than reaching the accessory ones, so the required recall is the primary signal.

Noise. When noise is present, we additionally report how many produced pairs draw their evidence from a non-gold passage, and how many of those still match a golden evidence. A high count signals an attributor that is distracted by plausible but irrelevant passages.

5.4.2 LLM-as-a-Judge lens: logical support

In the judge lens there is no covering filter: every produced pair is in scope, so its precision is computed over $\Pi_C(q)$, the same set as the nugget overall precision. The predicate *correct* is the verdict of the judge model of Section 5.2,

$$J(c, \hat{e}) \in \{ \text{supported}, \text{not_supported} \}, \quad (5.8)$$

and a pair is correct when $J(c, \hat{e}) = \text{supported}$. The judge lens therefore reports a single number, its precision, the fraction of produced pairs judged supported, obtained from Equation 5.4 over $\Pi_C(q)$ with this predicate.

Why the judge lens has no recall. The absence of a recall here is a property of the lens, not an omission. A recall needs a gold set of expected items at the denominator, and the judge has none: it reads each produced pair and rules **supported** or **not_supported**, but it never sees, and has no notion of, the facts a correct answer *should* have contained. Any fraction one could form would place the system’s own output at the denominator and would simply be the precision aggregated at the level of the claim, not a recall. One could of course borrow the nugget gold and define the judge’s recall as nugget coverage, but that would tie the judge back to the nuggets and destroy the very independence that makes it a useful second lens (Section 5.2). We therefore keep the judge lens to its precision alone.

5.4.3 Aggregation and the rationale for two lenses

All metrics are computed per question and then averaged across questions (macro-average), so that every question weighs equally regardless of how many claims or nuggets it has.

The two lenses are kept distinct on purpose, because each is strong exactly where the other is weak, as summarized in Table 5.1. The nugget lens is anchored to a human gold standard and is cheap to run, and it is free of *circularity*, that is, of the situation in which an attributor is graded by the very criterion it implements. The evidence it is scored against is annotated by hand, before and independently of any run, so neither attributor has a structural advantage over the other. A benchmark such as ALCE, by contrast, validates citations with an NLI model: there, an NLI attributor would be measured against a judge of its own kind, and the errors the two share would count as successes instead of being exposed. The nugget lens is, however, partial: its matched precision and its recall score only the claims that pass the covering filter, and they inherit that filter’s failure mode, a claim wrongly paired to a nugget about another fact. The judge lens rules on every produced pair and needs no annotation, but its verdicts are those of a model rather than of a human, and it is costly, one model call per pair. Read together, the two lenses say that a method scoring well under both produces citations that point to the passage a human annotator marked as the supporting evidence, and that an independent judge, reading the pair on its own, finds logically sufficient; a divergence between the two lenses, of the kind constructed in Section 5.3, is itself informative, and we exploit it in the analysis of Chapter 6.

	Nugget lens	LLM-as-a-judge lens
Correct pair	Span matches a golden evidence	Judge says supported
Precision	Correct / all pairs	Supported / all pairs
Matched precision	Correct / matched pairs	Not defined
Recall	Gold coverage (req / opt)	Not defined (no gold)
Ground truth	Human gold (nuggets)	None (model judgment)
Thresholds	Covering θ , evidence τ	None
Circularity	None	None (independent judge)
Cost	Cheap (similarity to gold)	Costly (one call per pair)
Main limitation	Mis-pairing in covering	Model-based verdict

Table 5.1: The two evaluation lenses and their complementary trade-offs.

The two lenses share the overall precision, computed over every produced pair. On top of it the nugget lens reports the matched precision, over the

matched pairs alone, and the recall, the coverage of the gold nuggets, split into required and optional with the required recall as the primary signal; the judge lens reports its precision alone, for the reason given in Section 5.4.2. These are the quantities reported throughout the experiments of Chapter 6.

5.5 The evaluation journey

It is worth closing the chapter where Chapter 3 ended. Figure 3.4 traced an answer to the set of pairs $\langle c, \hat{e} \rangle$ the pipeline produces. Evaluation picks up exactly there and forks into the two lenses, as Figure 5.4 shows. The nugget lens first applies the covering filter, keeping the claims that cover a nugget, then compares their evidence against the golden evidence with σ , and reports a precision over all pairs, a matched precision over the covering ones, and a recall (the coverage of the gold nuggets, split into required and optional). The judge lens, in parallel, applies no filter and sends every pair to the judge for a binary verdict, reporting its precision over all pairs. The same pipeline output is thus read twice, once against what a human annotated and once under an independent model, and the experiments of Chapter 6 compare the NLI and LLM attributors under both readings.

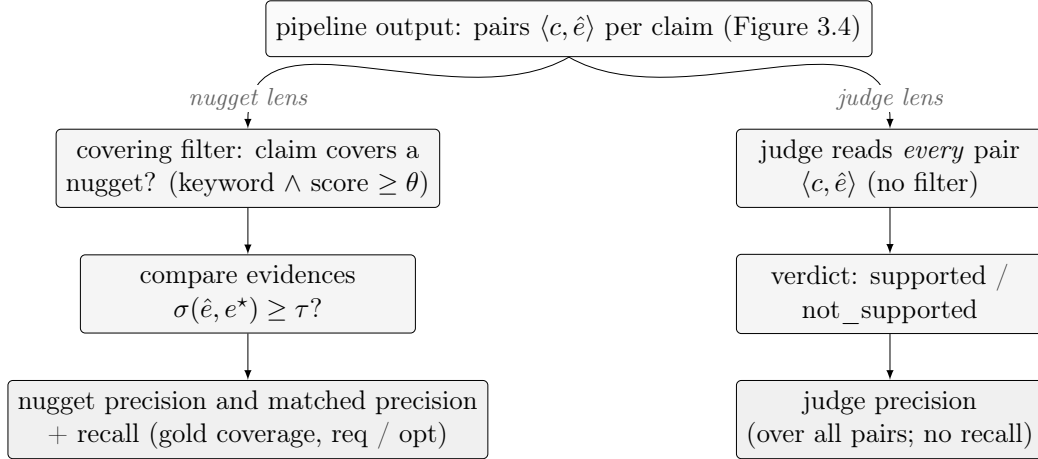


Figure 5.4: The evaluation journey, continuing the pipeline journey of Figure 3.4. The pairs produced by the pipeline are read by two lenses in parallel: the nugget lens (left) first filters the claims by the covering test, then compares the evidence of the matched claims against the human gold, reporting a precision over all pairs, a matched precision over the covering ones, and a recall; the judge lens (right) applies no filter and sends every pair to an independent model for a binary verdict.

Chapter 6

Experiments

The previous chapters fixed every moving part of this study except one. The pipeline of Chapter 3 defines how answers are generated, decomposed, and cited; the ALCE+ benchmark of Chapter 4 supplies the questions, the passages, the distractors, and the gold nuggets; the two lenses of Chapter 5 define how citation quality is measured. What remains open is the attribution function φ itself, and this chapter closes that question empirically. The comparison is guided by the question that motivates the thesis: how far can a local, efficient attributor go against a large language model, once both are dropped into the same pipeline and measured by the same gold?

The answer, in one sentence, is a trade-off rather than a winner: once its candidate set is gated by a cheap bi-encoder, the local cross-encoder becomes the most precise attributor in the comparison on every dataset, while the LLM covers more of the gold and produces more supported citations in absolute terms. The rest of the chapter establishes this result and dissects the mechanism behind it: why the plain cross-encoder collapses under retrieval noise, why the gating repairs it, and how the two evaluation lenses, agreeing where they should agree and disagreeing where they should disagree, certify that the measurement itself can be trusted.

6.1 Experimental setup

Datasets. All experiments run on the three subsets of ALCE+ introduced in Chapter 4, chosen because they stress an attributor in different ways. ASQA (28 questions) is factoid: ambiguous questions whose answers enumerate a handful of well-separated facts. ELI5 (30 questions) is long-form: explanatory answers whose claims are diffuse and whose evidence is spread across prose. QAMPARI (21 questions) is list-shaped: answers enumerate

many parallel entities, so a single question can produce dozens of structurally identical claims. A pattern that survives all three regimes is unlikely to be an artifact of any one of them.

Configurations. Five configurations are compared, obtained by crossing the attributor with the noise condition of Chapter 4: the LLM attributor (Claude Haiku 4.5) and the plain NLI attributor (the DeBERTa-v3-large cross-encoder), each on the clean oracle pool (*no noise*) and on the pool with injected distractors (*noise*); and the NLI attributor with the BGE sentence pre-filter enabled (*NLI+BGE*), evaluated in the noise condition, where the gating has something to defend against. There is no LLM+BGE configuration: the sentence pre-filter exists to shrink the candidate set of the cross-encoder, which scores sentence by sentence, whereas the LLM attributor reads whole passages and needs no such gate. Everything upstream of φ is held fixed: the same generator, the same decomposition step, the same oracle passages, the same thresholds ($\tau = 0.5$, $\text{top-}K = 3$).

Metrics and aggregation. Each run is scored under both lenses of Chapter 5. Under the nugget lens: precision (correct pairs over all produced pairs), matched precision (same numerator, restricted to the pairs of matched claims), and recall, the coverage of the gold nuggets, reported overall and on the required nuggets, which are the primary signal. Under the judge lens: precision, the fraction of produced pairs the independent judge deems supported, computed over the same denominator as the nugget precision and therefore directly comparable to it. All metrics are macro-averaged: computed per question and then averaged across questions, so that every question weighs equally regardless of how many claims its answer produces. Counts (pairs, correct pairs, supported pairs) are reported as raw sums, which do not depend on the aggregation. Section 6.7 reports cost and runtime.

A note on scale. With 21 to 30 questions per dataset, the large effects reported below (precision collapses of twenty to thirty points, noise-citation rates above fifty percent, zero correct pairs from noise) are unambiguous, but differences of a few points between configurations sit within ordinary run-to-run variability and are not over-interpreted. A related source of variability is the decomposition step, which is itself an LLM call and not deterministic: claim counts differ by less than two percent between runs of the same configuration, which does not affect any qualitative conclusion.

6.2 Main comparison

Table 6.1 reports the full matrix, five configurations by three datasets, and Table 6.2 the underlying counts. The single most informative reading of Table 6.1 is the head-to-head between the two noise-condition contenders, the LLM and the gated cross-encoder, since the noise condition is the realistic one: retrieval in the wild is never clean.

Dataset	Configuration	Nugget lens				Judge lens
		Prec.	Matched	Recall	Required	Prec.
ASQA	LLM (no noise)	44.1	73.0	54.0	75.7	90.5
	LLM (noise)	43.6	78.0	58.4	80.3	90.6
	NLI (no noise)	34.8	59.0	43.3	57.5	67.8
	NLI (noise)	25.0	44.6	48.0	66.6	45.7
	NLI+BGE (noise)	49.9	85.1	47.5	66.6	97.1
ELI5	LLM (no noise)	27.1	71.6	72.4	85.6	88.2
	LLM (noise)	27.4	69.5	73.6	85.6	89.4
	NLI (no noise)	22.2	53.9	65.1	79.4	66.0
	NLI (noise)	13.7	32.3	60.0	70.0	39.5
	NLI+BGE (noise)	34.4	77.1	62.2	73.3	91.5
QAMPARI	LLM (no noise)	51.5	74.0	76.5	91.9	90.3
	LLM (noise)	49.3	74.6	70.8	85.7	93.6
	NLI (no noise)	49.5	68.7	54.3	62.2	86.5
	NLI (noise)	32.4	49.8	51.4	59.4	59.4
	NLI+BGE (noise)	54.4	71.8	49.2	57.1	99.0

Table 6.1: Citation quality by attributor and condition, macro-averaged per question, in percent. Under the nugget lens, precision and matched precision share the numerator and differ only in the denominator; recall is the coverage of the gold nuggets. Under the judge lens, precision is computed over the same pairs as the nugget precision. Bold marks the best value per dataset among the two noise-condition contenders (LLM noise and NLI+BGE noise).

Dataset	Configuration	Claims	Pairs	Correct	Supported	Cit./claim
ASQA	LLM (no noise)	402	603	261	544	1.50
	LLM (noise)	411	619	262	565	1.51
	NLI (no noise)	417	469	155	299	1.12
	NLI (noise)	403	713	155	303	1.77
	NLI+BGE (noise)	407	334	163	323	0.82
ELI5	LLM (no noise)	520	723	188	635	1.39
	LLM (noise)	511	720	183	644	1.41
	NLI (no noise)	536	650	133	397	1.21
	NLI (noise)	487	973	128	379	2.00
	NLI+BGE (noise)	489	383	122	358	0.78
QAMPARI	LLM (no noise)	330	402	196	361	1.22
	LLM (noise)	315	354	176	331	1.12
	NLI (no noise)	333	265	126	225	0.80
	NLI (noise)	321	363	114	202	1.13
	NLI+BGE (noise)	317	208	113	205	0.66

Table 6.2: Raw counts per run: decomposed claims, produced claim-evidence pairs, pairs correct under the nugget lens, pairs supported under the judge, and citations per claim. Counts are sums and do not depend on the aggregation. Claim counts differ slightly between runs because the decomposition step is not deterministic.

Three regularities emerge. First, the gated cross-encoder is the most precise attributor in the field: NLI+BGE posts the best precision and the best judge precision on all three datasets, and the best matched precision on ASQA and ELI5, with QAMPARI essentially tied (71.8 against the LLM’s 74.6, a gap within the variability discussed in Section 6.1). Second, the LLM covers more of the gold everywhere: its recall exceeds the gated cross-encoder’s by roughly ten to twenty points per dataset (58.4 vs 47.5 on ASQA, 73.6 vs 62.2 on ELI5, 70.8 vs 49.2 on QAMPARI), with the same ordering on the required nuggets. Third, the LLM’s generosity is not padding: Table 6.2 shows that it produces far more supported citations in absolute terms (565 against 323 on ASQA, 644 against 358 on ELI5, 331 against 205 on QAMPARI), so its extra citations are, by the judge’s account, overwhelmingly good ones.

The comparison is therefore a genuine precision-recall trade-off, and it is consistent across question types: the gated local model is the choice when every citation must be right, the LLM when the answer must be covered. Neither dominates. What makes this result non-obvious is where the cross-encoder’s precision comes from, because without the gate the same model is not merely worse than the LLM, it is the worst attributor in the table. The

next two sections take this apart.

Two further observations frame what follows. The LLM is essentially flat between the clean and the noisy condition on every dataset (on ASQA its matched precision even rises slightly, 73.0 to 78.0, a movement within noise-level variability): injected distractors simply do not tempt it. The plain NLI attributor, by contrast, loses between eight and twenty-seven points across the precision metrics the moment distractors enter the pool. The noise condition is where the two families separate.

6.3 The effect of noise: the plain cross-encoder collapses

Table 6.3 isolates the noise behaviour of the NLI family. The numbers to read first are in the last three columns: how many distractor passages each configuration cites, how many claims cite at least one distractor, and how many of those citations turn out to be correct.

Dataset	Configuration	Noise passages	Claims citing noise	Correct from noise
ASQA	NLI (noise)	296	221 (54.8%)	0
ASQA	NLI+BGE (noise)	0	0 (0.0%)	0
ELI5	NLI (noise)	455	310 (63.7%)	0
ELI5	NLI+BGE (noise)	1	1 (0.2%)	0
QAMPARI	NLI (noise)	139	103 (32.1%)	0
QAMPARI	NLI+BGE (noise)	1	1 (0.3%)	0

Table 6.3: Noise usage of the NLI family, all-pairs scope: distractor passages cited, claims citing at least one distractor (share of all claims), and pairs from distractors that are correct under the nugget lens. The LLM attributor cites zero distractor passages on every dataset and is omitted. Not a single citation drawn from a distractor, in any run, covers a nugget or matches the gold.

The plain cross-encoder is severely distracted. On ASQA more than half of the claims (54.8%) cite at least one injected distractor; on ELI5 nearly two thirds (63.7%); on QAMPARI a third (32.1%). And these citations are pure loss: across the three datasets and 890 distractor citations, not one covers a nugget or matches the gold evidence. The judge concurs independently, and its verdict is visible in Table 6.1 as the collapse of the judge precision (67.8 to 45.7 on ASQA, 66.0 to 39.5 on ELI5, 86.5 to 59.4 on QAMPARI).

How the loss manifests differs slightly by dataset, and the difference is instructive. On ASQA the number of correct pairs is exactly unchanged between the clean and the noisy run (155 in both, Table 6.2): the distractors do not displace good citations, they pile roughly 250 additional wrong pairs on top of them, and precision falls purely because the denominator inflates. On QAMPARI the correct pairs also dip (126 to 114): with a fixed top- K per claim, distractor sentences occasionally push a good citation out of the returned set. On ELI5 the distraction rate is the highest of the three, consistent with the long-form regime offering the largest pool of candidate sentences to be wrong about, yet the loss follows the ASQA pattern rather than the QAMPARI one: the correct pairs are essentially unchanged between the clean and the noisy run (133 against 128) while the produced pairs balloon from 650 to 973, so once again the collapse is almost pure denominator inflation.

The LLM, by contrast, is immune: zero distractor passages cited on all three datasets. Its precision, matched precision, judge precision, and recall are statistically flat between conditions (Table 6.1). Whatever mechanism lets it decide that a topically adjacent but irrelevant passage does not support a claim, the entailment threshold of the cross-encoder does not replicate it. The failure is systematic, it grows with the amount of prose, and it is the single largest quality gap in the whole comparison.

6.4 The fix: gating the candidate set

The BGE pre-filter repairs the collapse almost entirely. Distractor citations drop from 296 to 0 on ASQA, from 455 to 1 on ELI5, from 139 to 1 on QAMPARI (Table 6.3); the judge precision rebounds from 45.7 to 97.1, from 39.5 to 91.5, from 59.4 to 99.0; and the gated model overtakes the clean-pool baseline itself (49.9 precision under noise against the plain model’s 34.8 without any noise at all on ASQA; 34.4 against 22.2 on ELI5; 54.4 against 49.5 on QAMPARI).

The recall cost of this surgery is remarkably small. On ASQA the required recall of the gated and the ungated model under noise is identical (66.6 in both). On QAMPARI it dips by two points (59.4 to 57.1). On ELI5 the gated model’s recall is actually higher than the ungated one’s (62.2 against 60.0 overall, 73.3 against 70.0 on the required nuggets): on the noisiest dataset, cleaning the candidate set frees the top- K slots that distractors were occupying and lets more gold evidence through. The gate does not trade coverage for precision; it removes citations that were contributing nothing.

It removes a great many of them. Citations per claim fall from 1.77 to

0.82 on ASQA, from 2.00 to 0.78 on ELI5, from 1.13 to 0.66 on QAMPARI (Table 6.2): the gate roughly halves the citation volume, and on ELI5 cuts it by more than sixty percent. That so large a cut costs so little recall is itself evidence about what was being cut: overwhelmingly, citations that covered no nugget and matched no gold. Section 6.5 verifies this directly by looking at the removed citations themselves.

The architectural reading is the central lesson of this chapter. The cross-encoder’s vulnerability is a candidate-set problem, not a scorer problem. DeBERTa applies an entailment threshold to each candidate sentence in isolation, and topically adjacent distractor sentences clear that per-item threshold; the bi-encoder imposes a relative, top- k gate on the candidate pool, and distractors lose a relative comparison against genuine evidence even when they would pass an absolute one. Gated, the cross-encoder never sees what it would have been fooled by. The cheapest component in the stack fixes the most powerful model’s failure mode, at no cost in coverage.

One honest caveat bounds this conclusion. The gated model under noise is more precise than the plain model on a clean pool (49.9 against 34.8 on ASQA, 54.4 against 49.5 on QAMPARI), and it also cites less on clean-equivalent volumes (0.82 against 1.12 citations per claim on ASQA). The gate therefore improves precision through two mechanisms at once: removing distractors, and cutting low-rank candidates that would have been wrong even without any distractors in the pool. The runs reported here cannot separate the two contributions; an NLI+BGE run on the clean pool would, by isolating the second mechanism, and is left as the natural control for future work. What the present data does establish is that the noise-removal mechanism alone accounts for the difference between collapse and best-in-field, since it is the only mechanism that distinguishes the noise condition.

6.5 Diagnosing the cross-encoder

The aggregate numbers say that the plain cross-encoder cites distractors; they do not say why, or which citations the gate removes. This section closes both gaps with two complementary probes: an error analysis of the distractor citations themselves, and a ranking probe that compares the top-1 and top-3 behaviour of each attributor. The mechanistic account of the underlying bias, at the level of the model’s internal computation, is developed in Appendix B; here the analysis stays at the level of observable behaviour.

6.5.1 What the cross-encoder wrongly entails

Inspecting the noisy QAMPARI run pair by pair reveals a sharply repetitive pattern: the same distractor sentences are entailed over and over, for many unrelated claims, and essentially all such pairs are rejected by the judge. Table 6.4 shows the most cited ones.

Distractor sentence (abridged)	Passage	Claims	Supported
“It has been adapted for film, television, and stage multiple times.”	<i>Murder on the Orient Express</i>	12	0
“Italian remains the most common language used in opera performance.”	<i>Italian opera</i>	10	0
“It was first performed on 24 December 1871 at the Khedivial Opera House in Cairo.”	<i>Aida</i>	8	0
“Italian opera is both the art of opera in Italy and opera in the Italian language.”	<i>Italian opera</i>	7	0

Table 6.4: The most frequently over-entailed distractor sentences in the noisy QAMPARI run: number of distinct claims each sentence was attached to, and how many of those pairs the judge deemed supported. The claims in question concern, among other things, football World Cup winners and mountains above eight thousand metres.

Two features of the pattern stand out. The first is the type of sentence the cross-encoder over-entails: generic, self-contained declaratives, definitional sentences and theme statements, often with a vague anaphoric subject and an unfalsifiable predicate (“*it* has been adapted . . . *multiple times*”). Such sentences read like facts, share topical vocabulary with everything, and entail nothing in particular; the cross-encoder mistakes this surface plausibility for support, attaching a sentence about opera performance to ten claims about eight-thousand-metre mountains, and a sentence about a novel’s adaptations to eleven claims about World Cup winners.

The second is the amplification mechanism, and it explains why QAMPARI suffers in this particular way. QAMPARI claims are lists of parallel, structurally identical facts. If a generic sentence clears the entailment threshold for one such claim, it clears it for all of them, because the claims are near-duplicates of each other; a single spurious entailment therefore replicates across an entire enumeration, which is exactly the twelve-fold and ten-fold repetition visible in Table 6.4. The list structure does not create the bias; it multiplies it.

This analysis also answers the question left open in Section 6.4: what does the gate remove? Precisely these sentences. The over-entailed distractors of Table 6.4 come from passages that are topically foreign to the question, and a relative ranking against the genuine evidence sentences eliminates them from the candidate set before the cross-encoder ever scores them. The error analysis and the candidate-set argument are two views of the same fact. Appendix B pursues the remaining question, why the cross-encoder assigns high entailment probability to such sentences at all, into the model itself.

6.5.2 Ranking as diagnosis: top-1 against top-3

The second probe exploits a structural difference between the two attributors. The NLI attributor returns up to $K = 3$ passages per claim ranked by entailment score, so restricting evaluation to the top-1 citation asks how good its *best* citation is; the LLM attributor returns an unranked set of citations with no scores, so its top-1 is an arbitrary member of the set. If the cross-encoder’s errors concentrate in the tail of its own ranking, its top-1 metrics should jump; if the LLM’s citations are exchangeable, its top-1 metrics should stay flat. Table 6.5 confirms both predictions.

Dataset	Configuration	Matched precision	Judge precision	Recall
		top-3 $\rightarrow$ top-1	top-3 $\rightarrow$ top-1	top-3 $\rightarrow$ top-1
ASQA	LLM (noise)	78.0 $\rightarrow$ 76.7	90.6 $\rightarrow$ 95.4	58.4 $\rightarrow$ 56.9
	NLI (noise)	44.6 $\rightarrow$ 63.0	45.7 $\rightarrow$ 69.7	48.0 $\rightarrow$ 45.6
	NLI+BGE (noise)	85.1 $\rightarrow$ 84.3	97.1 $\rightarrow$ 97.4	47.5 $\rightarrow$ 46.1
ELI5	LLM (noise)	69.5 $\rightarrow$ 75.3	89.4 $\rightarrow$ 92.4	73.6 $\rightarrow$ 70.2
	NLI (noise)	32.3 $\rightarrow$ 57.6	39.5 $\rightarrow$ 67.3	60.0 $\rightarrow$ 58.9
	NLI+BGE (noise)	77.1 $\rightarrow$ 80.1	91.5 $\rightarrow$ 91.7	62.2 $\rightarrow$ 62.2
QAMPARI	LLM (noise)	74.6 $\rightarrow$ 79.1	93.6 $\rightarrow$ 96.7	70.8 $\rightarrow$ 67.8
	NLI (noise)	49.8 $\rightarrow$ 58.3	59.4 $\rightarrow$ 73.0	51.4 $\rightarrow$ 48.2
	NLI+BGE (noise)	71.8 $\rightarrow$ 72.8	99.0 $\rightarrow$ 98.2	49.2 $\rightarrow$ 47.5

Table 6.5: Effect of restricting each claim to its single best citation (top-1) instead of up to three, macro-averaged. For the NLI attributor “best” means highest entailment score; the LLM attributor produces no scores, so its top-1 is not a ranked choice. The plain NLI attributor gains between fourteen and twenty-eight points of judge precision at top-1; the gated and the LLM attributor are essentially flat.

The plain cross-encoder’s top-1 jumps dramatically: on ASQA its judge precision rises from 45.7 to 69.7 and its matched precision from 44.6 to 63.0;

on QAMPARI from 59.4 to 73.0 and from 49.8 to 58.3; and on ELI5, the dataset where distraction is worst, the jump is the largest of all, from 39.5 to 67.3 and from 32.3 to 57.6. Its own entailment score, in other words, knows more than its citation set lets on: the highest-scored citation is usually a good one, and the errors, including the spurious entailments of Table 6.4, concentrate in the second and third ranks. The gated model is flat at top-1 (97.1 to 97.4 judge on ASQA) because the gate has already removed that tail, and the LLM is flat to mildly mixed (its judge precision rises a few points while its full precision falls, 43.6 to 38.7 on ASQA) because with no internal ranking there is no tail for top-1 to cut.

The ranking probe thus corroborates the error analysis with a purely quantitative instrument: the cross-encoder’s failure lives in the low-rank region of its own scoring, which is exactly the region a relative gate removes. It also shows that the top- K return policy, not the scorer alone, shapes the headline numbers, which is why the main results should be read jointly with this control.

6.6 The two lenses validate each other

Chapter 5 argued that the nugget lens and the judge lens fail in different ways and should therefore be read together. The experimental matrix bears this out with a pattern that doubles as a validation of the measurement itself.

On the weak configurations the lenses converge. For the plain NLI attributor under noise, the judge precision and the matched precision nearly coincide (45.7 against 44.6 on ASQA, 59.4 against 49.8 on QAMPARI, 39.5 against 32.3 on ELI5): when citations are genuinely bad, gold-matching and independent judgment agree that they are bad. On the strong configurations the lenses diverge widely: the LLM under noise scores 90.6 under the judge but 43.6 under the full nugget precision on ASQA. The divergence is not a contradiction but a measurement of gold incompleteness: the LLM cites many true claims whose facts the seven-odd nuggets of a question simply do not enumerate, and pairs of that kind can never count as correct under the nugget lens no matter how good the citation is. Appendix A shows the mechanism on a single real question, where the same run scores 28% under one lens and 81% under the other, with each individual discrepancy traceable either to an unlisted true claim or to a mis-paired one.

A subtler observation strengthens the case for the judge’s independence. The judge is itself an LLM, which raises the legitimate worry that it might systematically favour the LLM attributor, whose notion of support it plausibly shares. The results refute the strong form of this worry: the judge

awards its highest scores not to the LLM but to the gated cross-encoder, on every dataset (97.1, 91.5, 99.0 against the LLM’s 90.6, 89.4, 93.6). A judge that graded style rather than substance could not prefer the method that shares none of its architecture. The residual, weaker form of the worry, that the judge’s leniency might inflate all LLM-adjacent scores by some common offset, cannot be excluded by internal evidence and is listed among the limits in Section 6.8.

Taken together, the convergence on weak systems, the interpretable divergence on strong ones, and the judge’s preference for the non-LLM attributor indicate that the two lenses measure a real property of the citations rather than artifacts of their own construction. This is the sense in which the evaluation framework of Chapter 5 is a contribution and not merely an instrument.

6.7 Cost and runtime

The comparison so far concerns quality; the motivating question of the thesis also concerns efficiency, and the two attributors sit at opposite ends of that axis. The LLM attributor requires an API call per claim to a commercial model, with the attendant monetary cost, latency variability, and non-determinism; the NLI attributor is a single locally hosted cross-encoder, free per query, deterministic, and auditable. The BGE gate adds a bi-encoder pass that is cheap by construction.

Table 6.6 reports the wall-clock duration of each run. Two caveats apply: the figures are end-to-end and include the DeepSeek judge calls, which are common to all configurations, and they were collected on a single machine without controlling for load, so they are indicative rather than benchmarks.

Dataset	Configuration	Questions	Runtime	Per question
ASQA	LLM (no noise)	28	0h15m	31s
	LLM (noise)	28	0h15m	33s
	NLI (no noise)	28	2h32m	325s
	NLI (noise)	28	2h40m	344s
	NLI+BGE (noise)	28	0h57m	122s
ELI5	LLM (no noise)	30	0h17m	34s
	LLM (noise)	30	0h20m	40s
	NLI (no noise)	30	5h03m	608s
	NLI (noise)	30	3h36m	431s
	NLI+BGE (noise)	30	4h19m	517s
QAMPARI	LLM (no noise)	21	0h10m	29s
	LLM (noise)	21	0h09m	26s
	NLI (no noise)	21	1h19m	226s
	NLI (noise)	21	2h50m	486s
	NLI+BGE (noise)	21	0h37m	107s

Table 6.6: Wall-clock runtime per run, end to end, including judge calls. Figures are indicative: runs were not executed under controlled load.

Three readings. First, on this hardware the LLM runs complete in minutes while the NLI runs take hours: the cross-encoder scores every claim-sentence pair locally, and on a modest GPU that is the dominant cost. The efficiency argument for the local model is therefore not about wall-clock speed on arbitrary hardware but about the absence of per-query monetary cost, the determinism of the output, and independence from an external provider; on server-class hardware the picture would compress. Second, the gate also pays for itself computationally on two of the three datasets: by shrinking the candidate set it cuts the NLI runtime by roughly a factor of three to four (2h40m to 0h57m on ASQA, 2h50m to 0h37m on QAMPARI). On ELI5 the wall-clock figures behave erratically (the gated run was not faster than the ungated noisy one, 4h19m against 3h36m, and the clean run was slower still at 5h03m despite scoring fewer pairs than the noisy one), which the uncontrolled-load caveat above most plausibly explains and which is why these figures are read as indicative only. Third, the monetary cost of the whole campaign is modest, and its distribution is itself a result. Table 6.7 reports the API spend, measured as account-balance deltas over each block of runs. The six LLM-attributor runs cost \$5.81 of Claude credit, of which ELI5, with the longest answers, is the most expensive dataset (\$2.55, against \$1.92 for ASQA and \$1.34 for QAMPARI), plus \$0.39 of DeepSeek credit for

the judge. The nine NLI-family runs cost \$1.25 of Claude credit, which pays only for the shared generation and decomposition steps, since the attribution itself runs locally at zero marginal cost, plus \$0.44 for the judge. Per question, the LLM configuration costs roughly four cents against under one cent for the NLI family, and the entire fifteen-run matrix, judging included, cost \$7.89. Two conclusions follow. The judge lens is cheap enough (\$0.83 in total) to be used routinely rather than reserved for spot checks. And the recurring cost that separates the two attributors is exactly the attribution call: everything else in the pipeline is shared infrastructure.

Block	Claude (USD)	DeepSeek (USD)	Total (USD)
LLM attributor, 6 runs	5.81	0.39	6.20
NLI family, 9 runs	1.25	0.44	1.69
Full matrix, 15 runs	7.06	0.83	7.89

Table 6.7: API spend per block of runs, measured as account-balance deltas. Claude credit covers generation, decomposition, and, in the LLM block only, attribution; DeepSeek credit covers the judge. In the NLI family the attribution step runs locally and incurs no API cost.

6.8 Limits and threats to validity

Six limits bound the conclusions of this chapter, none of which, in our assessment, threatens the qualitative picture.

Sample size. With 21 to 30 questions per dataset, only the large effects are beyond doubt; differences of a few points (the QAMPARI matched-precision tie, the small recall dips) are within run-to-run variability and are not interpreted.

Stochastic decomposition. The decomposition step is an LLM call, so the claim sets of different runs differ by up to two percent. The attributors are therefore compared on nearly, not exactly, identical claims. Caching the decomposition would remove this confound and is recommended for any follow-up; no conclusion of this chapter rests on a difference small enough to be affected.

The two mechanisms of the gate. As discussed in Section 6.4, the gate improves precision both by removing distractors and by cutting low-rank candidates; the runs reported here cannot apportion the gain between the two. The missing control is an NLI+BGE run on the clean pool.

Judge independence. The judge is an LLM and might share biases with

the LLM attributor. The strong form of the worry is refuted by the judge’s preference for the gated cross-encoder (Section 6.6); the weak form, a uniform leniency offset, cannot be excluded without a second, architecturally distinct judge.

Aggregation. All percentages are macro-averages per question, consistent with Chapter 5. Pooled (per-pair) aggregates were also computed and tell the same qualitative story; where the two differ materially (list-shaped QAMPARI, where question sizes vary most), the macro figures are the ones reported, since letting a few long enumerations dominate the average is precisely what macro-averaging exists to prevent.

Oracle setting. All runs use oracle passages with injected distractors, which isolates the attribution function but idealises retrieval. The noise condition is a step toward realism, not a substitute for it; end-to-end retrieval remains future work, and the candidate-set lesson of Section 6.4 is expected to matter more there, not less.

The top-1 control of Section 6.5.2 is reported in the main text rather than an appendix precisely because it is diagnostic and not merely defensive; the full per-run top-1 tables are available alongside the released evaluation outputs.

Chapter 7

Conclusions

This thesis set out to answer a deployment question with a measurement problem inside it: how far can a local, efficient attributor go against a large language model at assigning citations to the atomic claims of a generated answer, and how would one even tell? Concretely, the local attributor is a **DeBERTa-v3-large** NLI cross-encoder, an encoder-only model of 435 million parameters run on a laptop without a dedicated GPU and optionally gated by a **BGE** bi-encoder pre-filter, and the large language model is **Claude Haiku 4.5**, reached through a commercial API; the pipeline around them, and the answer they annotate, are held identical, so that the only thing varying across the comparison is the attribution function itself. Answering the question required building the measuring instrument before running the comparison, and the instrument turned out to be as much a contribution as the answer. This chapter summarizes what was built and what was found, states the answer to the motivating question, draws the boundaries of what the results can support, and lays out the work that follows.

7.1 Summary of contributions

The thesis makes three contributions, in the order in which the chapters build them.

The first is **ALCE+**, a claim-level benchmark for citation attribution (Chapter 4). Existing benchmarks lack claim-level ground truth; **ALCE+** supplies it in the form of hand-annotated atomic facts, the nuggets, split into required and optional, each carrying its own golden evidence, and complemented by injected distractor passages that turn robustness to retrieval noise into a measurable property rather than an anecdote.

The second is a **two-lens evaluation framework** (Chapter 5): a nugget

lens that matches the pipeline’s claims against the gold through a two-stage covering and scoring procedure, and an independent LLM-as-judge lens that rules on each claim-evidence pair without ever consulting the gold. The framework validates itself in the data: the two lenses converge on weak configurations, where citations are genuinely bad and both instruments say so, and diverge interpretably on strong ones, where the divergence measures the incompleteness of any finite gold rather than a disagreement about quality. A further check falls out of the results: the judge, itself an LLM, awards its highest scores to the non-LLM attributor, which is difficult to reconcile with the worry that it merely favours its own kind.

The third is the **empirical comparison and its diagnosis** (Chapter 6 and Appendix B). Across three question regimes, the comparison localizes exactly where the two attribution families differ: the plain cross-encoder collapses under retrieval noise, citing distractors for a third to two thirds of the claims, with 890 distractor citations across the runs and not a single one matching the gold, while the LLM is immune; a cheap bi-encoder gate on the candidate set repairs the collapse almost entirely (296 distractor citations to zero on ASQA, 455 to one on ELI5) at nearly no cost in coverage; and the interpretability case study traces the collapse to a structural bias of the cross-encoder, which ratifies surface co-occurrence schemas rather than verifying support, a diagnosis reached by refuting one hypothesis (attestation bias) and corroborating another (schema matching) with attention analysis, feature attribution and a causal intervention. The entire experimental campaign, judging included, cost \$7.89 of API credit.

7.2 The answer to the research question

The comparison yields a trade-off, not a winner, and the trade-off is consistent across all three datasets. Once its candidate set is gated, the local cross-encoder is the most precise attributor in the field: it posts the best precision and the best judge precision on every dataset, at zero marginal cost per query, deterministically, and without an external provider. The LLM covers more: its recall of the gold facts exceeds the gated model’s by ten to twenty points everywhere, and its greater citation volume is not padding, since it produces far more judge-supported citations in absolute terms.

The deployment reading follows directly. Where every citation must be right and volume is secondary, in audit-like settings, the gated local model is the better attributor, at under one cent of shared infrastructure cost per question. Where the answer must be covered and a reader tolerates the occasional loose citation, the LLM is, at roughly four cents per question in

this setup. Neither dominates, and the choice is now quantified rather than guessed.

The architectural lesson is the finding this thesis would keep if it could keep only one. The cross-encoder’s fragility is a candidate-set problem, not a scorer problem: a per-item, absolute entailment threshold is unsafe for a model that ratifies surface schemas, because schema-instantiating distractors clear it; a relative, top- k gate eliminates them before the scorer ever sees them. The cheapest component in the stack fixes the failure mode of the most powerful model in it, at no cost in coverage. How far a local, efficient attributor can go, then, depends less on the scorer than on what the scorer is allowed to see.

7.3 Limitations

Five boundaries qualify these conclusions, beyond the run-level threats already discussed in Section 6.8.

Oracle retrieval. All experiments hold retrieval fixed and correct, isolating the attribution function; the noise condition is a controlled step toward realism, not a substitute for end-to-end retrieval, where pool sizes and error modes differ.

One representative per family. The comparison instantiates the LLM family with Claude Haiku 4.5 and the NLI family with DeBERTa-v3-large. The results are about these representatives; the candidate-set argument is architectural and should transfer, but the magnitudes need not.

Gold incompleteness. No finite nugget set enumerates every true claim of an answer; the thesis measures this effect rather than suffering it silently, through the divergence of the two lenses, but the nugget-lens numbers remain lower bounds on citation quality.

Single-annotator gold. The nuggets are hand-annotated, which is what makes them independent of any model, but inter-annotator agreement has not yet been quantified: the gold’s own variability is currently unmeasured, and Section 7.4 makes measuring it the first item of future work.

Lexical component in the matching. The claim-to-nugget covering uses keyword anchoring, which is transparent and inspectable but lexical: a claim can capture a required nugget by sharing its keyword alone, as the mis-pairing documented in Appendix A shows on a real question. The evaluation inherits a small, identifiable error mode from this design choice.

7.4 Future directions

The future work divides into two threads, consolidating the instrument and hardening the attributor, plus the completion of the experimental matrix. Each item is anchored to a specific limitation or finding above.

Consolidating ALCE+ as a measuring instrument. Two steps would harden the gold. First, *quantifying annotation agreement*: the nuggets should be annotated independently by multiple annotators and the agreement and its variability measured and reported, turning the single-annotator limitation into a characterized property of the benchmark. Second, *matching beyond keyword overlap*: the covering stage should be made robust to purely lexical captures, for instance by requiring entity-level or semantic compatibility between claim and nugget in addition to the keyword anchor, so that a wrong claim can no longer win a required nugget by sharing its words alone.

Hardening the NLI attributor. The interpretability results of Appendix B motivate two concrete defenses, both of which act on exactly the failure modes the case study isolated. First, an *entity-guided threshold*: extracting the entities of premise and hypothesis with a generalist recognizer such as GLiNER [24] and modulating the entailment threshold on their actual overlap would directly counter the two most striking behaviours of the bias, the schema that fires with no shared entities at all and the single lexical bridge that resurrects a killed inference. Second, *premise-hypothesis normalization* before the forward pass: resolving coreference, making entities explicit, and normalizing register would remove the structural false positives that the case study traced to surface form, from the anaphoric-subject corpus sentences to the capitalization flip that moves the entailment probability by two thirds of its range. Behind both stands the deeper program the appendix opens: identifying the schema circuit with per-head analysis and path patching on length-matched pairs, and then fine-tuning on schema-breaking contrast pairs, with the aim of recovering the gated configuration’s precision without the gate, fixing the attributor at the source. Understanding *why* the model entails these pairs is the precondition for all of it.

Completing and extending the comparison. Three runs and two extensions remain. The NLI+BGE configuration on the clean pool would separate the gate’s two entangled mechanisms, noise removal and low-rank pruning; caching the decomposition would remove the last stochastic confound between attributors; and a second, architecturally distinct judge would close

the residual worry about judge leniency. Beyond the current setup, end-to-end retrieval is the natural next regime, and the candidate-set lesson is expected to matter more there, not less; and the cost analysis suggests a *hybrid cascade* worth testing: since the cross-encoder’s top-ranked citation is almost always good, a system could attribute every claim locally and escalate to the LLM only the claims whose entailment score falls below confidence, retaining most of the precision at a fraction of the cost.

The synthesis is the roadmap in one sentence: consolidate ALCE+ as a metric, by quantifying its agreement and hardening its matching, and strengthen the NLI attributor, by guarding its threshold with entities and normalizing its inputs; only then does the comparison between NLI and LLM attribution become definitive. The measurements of this thesis draw the map; the job is not finished.

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Appendix A

A Complete Run on a Real Question

Chapter 5 builds its intuition on a small, hand-constructed running example. This appendix complements it with a complete, unedited run of the system on a real benchmark question, so that every stage of the pipeline of Chapter 3 and every quantity of Chapter 5 can be read on genuine output. The question is item 23 of the ASQA gold-nugget split, “*What caused the fall of the Roman Empire?*”, which the benchmark equips with five passages and seven nuggets, four of them required and three optional. The attributor is the LLM method (Claude Haiku 4.5); noise injection is off, the support threshold is $\tau = 0.5$, at most $K = 3$ passages are returned per claim, and the sentence pre-filter is disabled, so this is the plain LLM configuration (Figure A.1).

The run is what makes the two lenses of Chapter 5 concrete: they diverge sharply here, an overall nugget precision of 28% against a judge precision of 81%, and the reason is not that the citations are wrong but that the gold does not enumerate every true claim of the answer and that one claim is mis-paired by the covering filter. The rest of the appendix follows the pipeline stage by stage and returns to this divergence at the end.

Impostazioni modello & retrieval

MODELLO LLM: claude-haiku-4-5-20251001

METODO RETRIEVAL: LLM

NLI THRESHOLD — 0.5

TOP-K PASSAGES — 3

PRE-FILTER FRASI (BGE) — OFF

NLI su tutte le frasi (più lento, più preciso)

1 Query

DATASET — ASQA_GOLD_NUGGETS.JSON

[23] What caused the fall of the Roman Empire?

Q: What caused the fall of the Roman Empire?

5 passages disponibili 7 nuggets · Modello: claude-haiku-4-5-20251001 · Noise OFF Cambia dataset

Figure A.1: Model and retrieval settings, and the selected question (ASQA item 23), with five passages and seven nuggets available.

A.1 Generate

The generation step conditions Haiku on the question and the five passages and returns a single paragraph of 134 words (Figure A.2):

Historians have identified multiple interconnected causes for the fall of the Western Roman Empire rather than a single factor. Military overextension, currency debasement, political instability, and overreliance on slave labor all contributed to the empire’s decline. However, the most immediate and visible pressure came from barbarian invasions, particularly from Germanic tribes and the Huns. The Hunnic invasions under leaders like Attila were especially devastating, forcing Germanic peoples such as the Visigoths and Vandals into Roman territory and destabilizing the empire’s borders from within. This broader period of widespread migrations, known as the Migration Period, occurred around 300–700 AD and fundamentally reshaped Europe’s ethnic and political landscape. The process culminated in 476 AD when the Germanic chieftain Odoacer deposed Romulus Augustulus, the last Western Roman emperor, marking the traditional end of the Western Roman Empire.

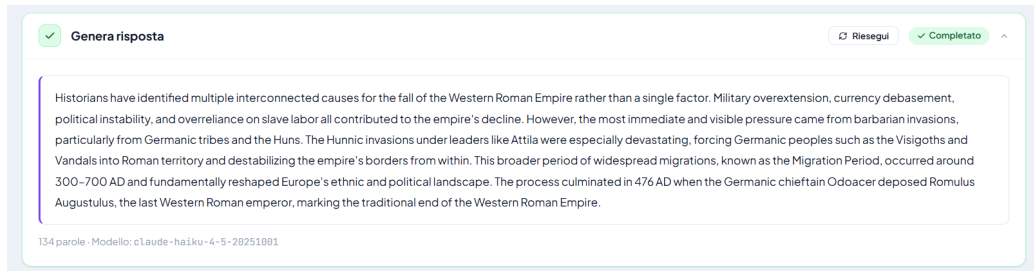


Figure A.2: Generate: the answer produced by Claude Haiku 4.5.

A.2 Decompose

The decomposition step rewrites the paragraph into nineteen atomic claims, each a single self-contained fact (Figures A.3 and A.4):

1. The fall of the Western Roman Empire had multiple interconnected causes rather than a single factor.
2. Military overextension contributed to the Western Roman Empire's decline.
3. Currency debasement contributed to the Western Roman Empire's decline.
4. Political instability contributed to the Western Roman Empire's decline.
5. Overreliance on slave labor contributed to the Western Roman Empire's decline.
6. Barbarian invasions were the most immediate and visible pressure on the Western Roman Empire.
7. Germanic tribes invaded the Western Roman Empire.
8. The Huns invaded the Western Roman Empire.
9. Attila led Hunnic invasions against the Western Roman Empire.
10. The Hunnic invasions were especially devastating to the Western Roman Empire.
11. The Visigoths were forced into Roman territory by Hunnic invasions.
12. The Vandals were forced into Roman territory by Hunnic invasions.
13. Barbarian invasions destabilized the Western Roman Empire's borders from within.
14. The Migration Period occurred around 300–700 AD.
15. The Migration Period fundamentally reshaped Europe's ethnic and political landscape.

16. Odoacer was a Germanic chieftain.
17. Odoacer deposed Romulus Augustulus in 476 AD.
18. Romulus Augustulus was the last Western Roman emperor.
19. The traditional end of the Western Roman Empire was marked in 476 AD.

These nineteen claims are the unit of everything that follows: each is attributed to passages independently, and each is the thing the nugget gold is matched against.

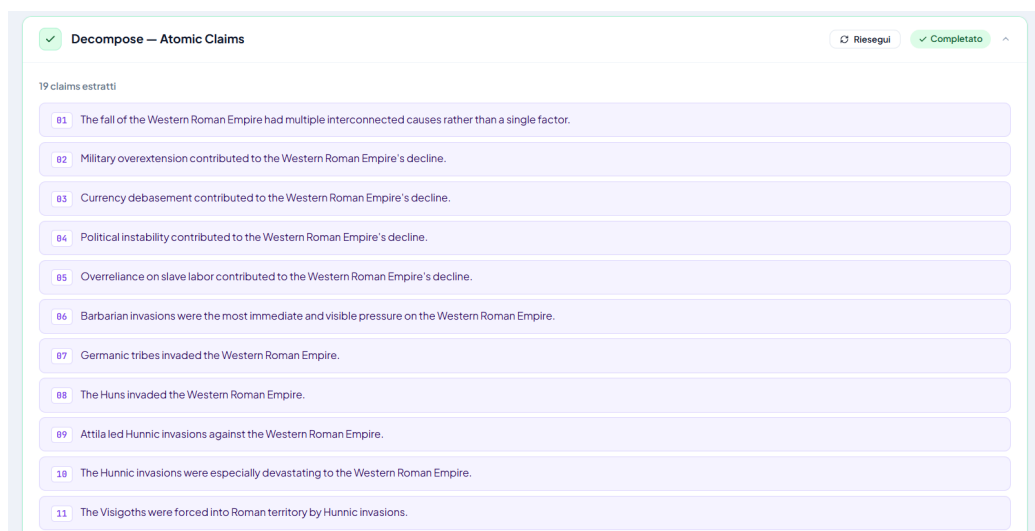


Figure A.3: Decompose: the first eleven atomic claims.

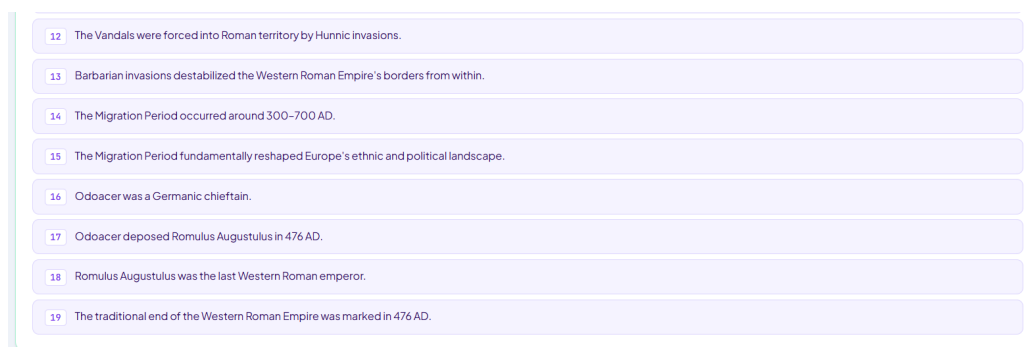


Figure A.4: Decompose: the remaining atomic claims.

A.3 Attribute and structural matching

The attribution step runs the LLM method on each of the nineteen claims and returns, for each, the passages that support it together with the sentence that constitutes the evidence; every claim receives at least one citation. This is where the citations are attached, before any evaluation takes place.

Two things share this stage in the interface, and it is worth keeping them distinct. The first is the attribution just described, the output of the function φ of Chapter 3. The second is a *preview* of the Stage-1 covering of Section 5.1: the structural claim-to-nugget matching is run here, ahead of the scored evaluation, so that the coverage of the gold can be inspected while the citations are still in view (Figures A.5 and A.6). The scored metrics themselves are computed later, in the Evaluate stage; what is shown here is only which claims cover which nuggets and with what covering score.

Table A.1 reports that preview. Of the seven nuggets, four are reached (three required, one optional) and three are left uncovered. Every covering score clears the covering threshold $\theta = 0.6$, in two cases only barely, at 0.61.

Nugget	Type	Covering claims (covering score)	Status
n0a	required	c17 (0.95), c19 (0.81)	covered
n0b	required	c18 (0.96)	covered
n0c	required	<i>none</i>	not covered
n1	required	c04 (0.72), c02 (0.72), c13 (0.61), c06 (0.61)	covered
n2	optional	c08 (0.70)	covered
n3a	optional	<i>none</i>	not covered
n3b	optional	<i>none</i>	not covered

Table A.1: Structural claim-to-nugget matching for the run. Nuggets are identified as in the benchmark; *cn* refers to the claim of that number in the decomposition. The two uncovered optional nuggets (**n3a**, **n3b**) concern the survival of the Eastern Empire until 1453, a fact the answer never states; the uncovered required nugget **n0c** is a matching artifact discussed below.

Retrieve — Matching claims — passaggio

19/19 claims supportati

Nugget matching: Gold (required) Silver (optional)

n0a REQUIRED 2 claims
The Western Roman Empire fell in 476 AD.
Keywords: 476 AD

✓ Odoacer deposed Romulus Augustulus in 476 AD. 0.95 2 fonti ▼

✓ The traditional end of the Western Roman Empire was marked in 476 AD. 0.81 2 fonti ▼

n0b REQUIRED 1 claim
Romulus Augustulus was the last emperor of the Western Roman Empire.
Keywords: Romulus Augustulus

✓ Romulus Augustulus was the last Western Roman emperor. 0.96 2 fonti ▼

n0c REQUIRED non coperto
Romulus Augustulus was deposed by Odoacer.
Keywords: Odoacer

Figure A.5: Structural matching: the required nuggets n0a and n0b, covered, and n0c, left uncovered.

n1 REQUIRED 4 claims
Causes of the fall of the Western Roman Empire included military overextension, political instability, and barbarian invasions.
Keywords: military overextension political instability barbarian invasions

✓ Political instability contributed to the Western Roman Empire's decline. 0.72 2 fonti ▼

✓ Military overextension contributed to the Western Roman Empire's decline. 0.72 1 fonte ▼

✓ Barbarian invasions destabilized the Western Roman Empire's borders from within. 0.61 3 fonti ▼

✓ Barbarian invasions were the most immediate and visible pressure on the Western Roman Empire. 0.61 3 fonti ▼

n2 OPTIONAL 1 claim
The Huns pushed Germanic peoples into Roman territory, destabilizing the empire.
Keywords: Huns Attila Germanic peoples

✓ The Huns invaded the Western Roman Empire. 0.70 3 fonti ▼

n3a OPTIONAL non coperto
The Eastern Roman Empire survived until 1453.
Keywords: Byzantine Empire 1453

n3b OPTIONAL non coperto
Constantinople fell to the Ottoman Empire in 1453.
Keywords: Constantinople Ottoman

Figure A.6: Structural matching: the required nugget n1 and the optional nuggets n2, n3a, n3b.

The uncovered required nugget n0c is instructive. Its fact is “*Romulus Augustulus was deposed by Odoacer*”, with keyword *Odoacer*, and the answer states exactly this in claim 17, “*Odoacer deposed Romulus Augustulus in 476 AD*”. The claim, however, was matched to n0a (“*The Western Roman Empire fell in 476 AD*”) on the shared keyword *476 AD*, at a covering score of 0.95, and did not also land on n0c. The fact is therefore stated and correctly cited

by the system, yet the gold records it as missed: the loss is in the pairing, not in the answer. This is the covering failure mode named in Section 5.2, seen on real output, and it is the kind of error the judge lens, which never consults a nugget, does not make.

Eleven of the nineteen claims match no nugget at all (Figure A.7). They are not errors: claims such as “*Currency debasement contributed to the decline*” or “*The Migration Period occurred around 300–700 AD*” are true and cited, but the seven nuggets simply do not enumerate them. These claims still receive citations, and their pairs still enter the overall denominator, a fact that matters for the precision below.

CLAIMS SENZA NUGGET (11)	
✓ The fall of the Western Roman Empire had multiple interconnected causes rather than a single factor.	3 fonti ▼
✓ Currency debasement contributed to the Western Roman Empire's decline.	1 fonte ▼
✓ Overreliance on slave labor contributed to the Western Roman Empire's decline.	1 fonte ▼
✓ Germanic tribes invaded the Western Roman Empire.	3 fonti ▼
✓ Attila led Hunnic invasions against the Western Roman Empire.	1 fonte ▼
✓ The Hunnic invasions were especially devastating to the Western Roman Empire.	3 fonti ▼
✓ The Visigoths were forced into Roman territory by Hunnic invasions.	1 fonte ▼
✓ The Vandals were forced into Roman territory by Hunnic invasions.	1 fonte ▼
✓ The Migration Period occurred around 300–700 AD.	1 fonte ▼
✓ The Migration Period fundamentally reshaped Europe's ethnic and political landscape.	1 fonte ▼
✓ Odoacer was a Germanic chieftain.	2 fonti ▼

Figure A.7: The eleven claims that match no nugget, each still carrying its attached citations.

Each citation can be inspected down to the evidence span (Figure A.8). For claim 17, for instance, the two attached passages are *Fall of the Western Roman Empire*, with the sentence “*The date most often cited for the ‘fall’ is 476 AD, when Romulus Augustulus . . . was deposed by the Germanic chieftain Odoacer*”, and *Odoacer*, with the sentence “*Odoacer . . . became the first King of Italy after deposing Romulus Augustulus . . . on September 4, 476 AD*”. Both are exact supports for the claim.

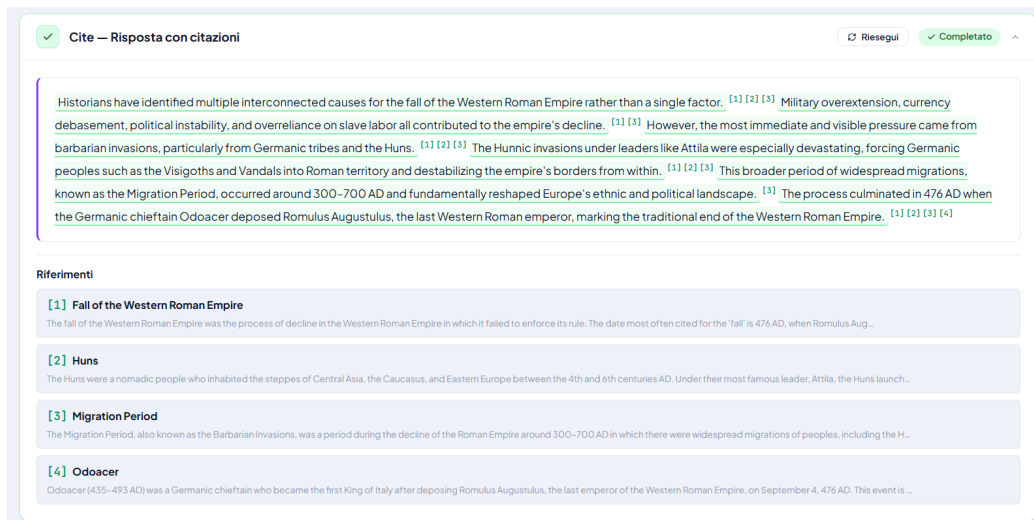


Figure A.9: Cite: the answer rendered with inline references [1] to [4] and the reference list.

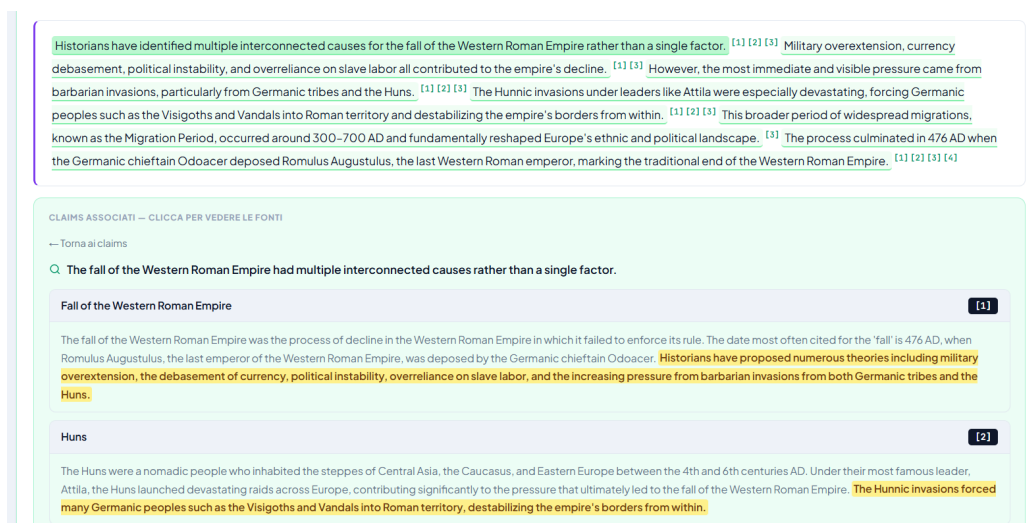


Figure A.10: Cite: clicking a highlighted sentence reveals its claims and, for each, the cited passages with the supporting span highlighted.

A.5 Evaluate: the two lenses

The evaluation stage scores the run under both lenses of Chapter 5 over the same thirty-six produced pairs. Table A.2 collects the numbers, and Figures A.11 and A.12 show them as reported by the system.

Lens	Metric	Value
Nugget	matched precision	10/18 = 56%
Nugget	overall precision	10/36 = 28%
Nugget	recall (coverage)	4/7 = 57%
Nugget	required	3/4 = 75%
Nugget	optional	1/3 = 33%
Judge	precision (supported)	29/36 = 81%

Table A.2: The two lenses on the run. The nugget precisions share the numerator (ten correct pairs) and differ only in the denominator, the thirty-six produced pairs against the eighteen pairs of the eight matched claims. The judge rules on all thirty-six pairs, finding twenty-nine supported and seven not.

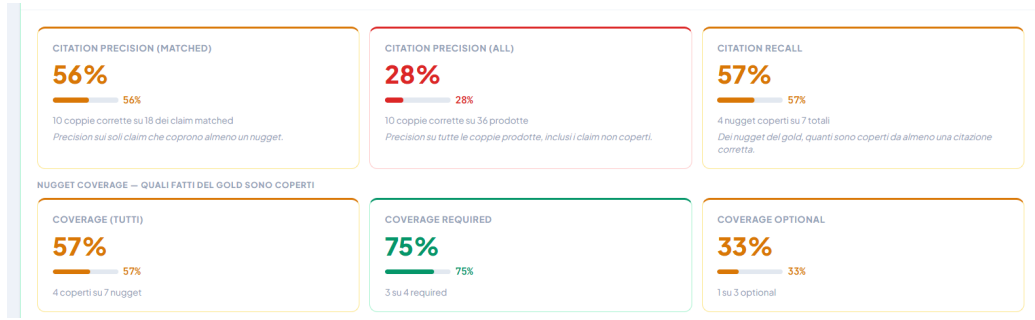


Figure A.11: Evaluate, nugget lens: the two precisions, the recall, and the nugget coverage split into required and optional.

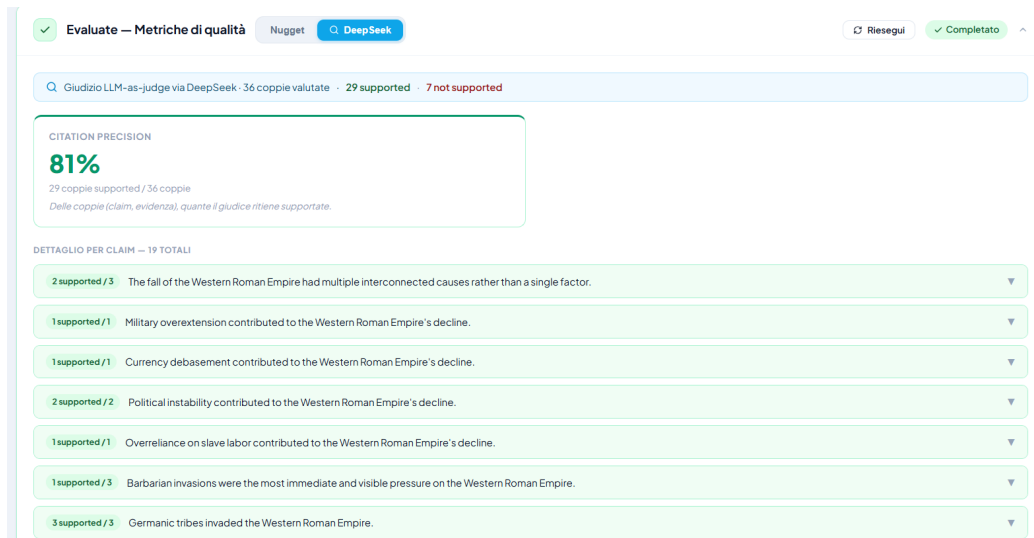


Figure A.12: Evaluate, judge lens: DeepSeek precision over the produced pairs, with the per-claim breakdown.

The two lenses disagree by a wide margin, and the run shows why. The overall nugget precision of 28% is held down almost entirely by the denominator: eighteen of the thirty-six pairs belong to the eleven claims that match no nugget, so they can never count as correct in this lens regardless of how good the citation is, and eight of the eighteen matched pairs do not clear the evidence threshold. The matched precision of 56% removes the first effect by restricting to matched claims, and already climbs halfway to the judge. The judge, reading each pair against its own cited evidence, credits twenty-nine of the thirty-six, including many pairs from the unmatched claims that are true and correctly supported but invisible to a gold that does not list their fact.

On top of this gold-incompleteness sits the mis-pairing of claim 17 discussed above: it costs a required nugget (n0c), pulling the required coverage down to 3/4 although the answer does contain the fact, and it lowers the nugget precision without any real fault in the citation. Both effects, the unlisted true claims and the mis-paired one, are exactly the failure modes for which Chapter 5 introduces the judge lens, and both are visible on a single real question here.

The judge's per-pair reasoning makes the complementarity concrete (Figure A.13). For claim 1, "*The fall had multiple interconnected causes rather than a single factor*", the judge marks two of its three citations supported and one not: the *Fall of the Western Roman Empire* passage, which lists several causes at once, and the *Migration Period* passage, which frames the

period as both cause and effect, are read as supporting the claim of multiple interconnected causes, while the *Huns* passage, which names only the Hunnic invasions, is read as not supporting a claim about *multiple* causes. The verdict is a judgment about logical support between a claim and a span, formed without any reference to the gold, and it is precisely this independence that lets the judge lens stand as a check on the nugget lens rather than a second view of it.



Figure A.13: Judge lens: the per-pair verdicts and DeepSeek rationales for the first claim, two supported and one not supported.

Appendix B

Inside the Entailment Bias of the Cross-Encoder

The error analysis of Chapter 6 established *that* the NLI attributor systematically entails sentences that support nothing, and characterized the sentences it favours. This appendix pursues the question one level deeper: *why* does the DeBERTa cross-encoder assign high entailment probability to such pairs? The investigation proceeds in three movements: a battery of minimal-pair probes that isolates the phenomenon outside the pipeline; a case study that puts one documented bias pair under three instruments, attention flow, Integrated Gradients, and activation patching, to arbitrate between two competing hypotheses; and a causal intervention that localizes where the bias signal travels through the network. The verdict, stated upfront, is provisional but directional: the first hypothesis, attestation bias, is refuted; the second, surface schema matching, is consistent with every probe and with the causal evidence, though the responsible circuit is not yet fully identified.

B.1 Background on mechanistic interpretability

The first thing we need to do, is to assess the theory around the techniques that are used in this appendix:

Attention flow. Raw attention weights are an unreliable measure of what influences a Transformer’s prediction, because information mixes across layers. Attention flow and attention rollout [1] correct for this by propagating attention through the layer stack, treating it as a flow network, and yield a more faithful picture of which input tokens actually reach the prediction.

Integrated Gradients. Integrated Gradients [20] is an axiomatic feature attribution method: the importance of each input feature is the path integral of the model’s gradients along the straight line from a neutral baseline input to the actual input. Unlike plain gradients, it satisfies sensitivity and implementation-invariance axioms, making scores comparable across inputs.

Activation patching. Where the previous techniques are correlational, activation patching [12] is causal: the internal activations of the model on one input are selectively replaced with those recorded on a different input, and the effect on the output is measured. If patching a specific component transfers the behavior from one input to the other, that component causally mediates the behavior. This is the technique that allows Appendix B to probe where in the model the entailment bias travels.

B.2 From corpus errors to isolated cases

Section 6.5.1 showed the corpus-scale face of the phenomenon on QAMPARI: generic, definitional distractor sentences attached to ten or twelve parallel claims each, with the judge rejecting essentially every such pair. The pattern is not a list-shaped quirk: it replicates on the long-form ELI5 runs, where the amplification works across questions rather than within enumerations. Table B.1 shows the most over-entailed sentences of the noisy ELI5 run: a theme statement about nuclear waste storage is attached to twenty-seven distinct claims across four different questions, definitional sentences with anaphoric subjects (“*It* is an organic chemical . . .”) and generic measurement clauses (“*This* is usually measured using the consumer price index”) are attached to eleven each, and, most tellingly, non-propositional form text from a scholarship application (“DETAILS OF OTHER FAMILY MEMBERS . . .”) is entailed for ten claims. The judge rejects nearly all of these pairs.

Over-entailed sentence (abridged)	Claims	Questions	Supported
“The long-term storage and disposal of nuclear waste remains one of the most significant challenges. . .”	27	4	1
“It is an organic chemical of the catecholamine and phenethylamine families.”	11	2	0
“This is usually measured using the consumer price index (CPI).”	11	2	0
“DETAILS OF OTHER FAMILY MEMBERS (grandfather, sisters, etc) SUPPORTED BY PARENTS. . .”	10	2	0
“Contrary to the popular ‘tongue map’, which suggested that different regions of the tongue. . .”	10	3	0

Table B.1: The most over-entailed sentences of the noisy ELI5 run: number of distinct claims each sentence was cited for, number of distinct questions those claims belong to, and how many of the pairs the judge deemed supported. The pattern of Section 6.5.1 replicates on long-form prose, with cross-question rather than intra-list amplification, and extends to text with no propositional content at all.

Corpus data of this kind establishes the phenomenon but confounds many variables at once: passage topic, sentence form, claim structure, and the surrounding candidate set all vary together. To dissect the mechanism, the investigation moves to isolated pairs, scored directly by the same checkpoint used in the pipeline (`cross-encoder/nli-deberta-v3-large`, a 24-layer model fine-tuned on SNLI, MultiNLI, FEVER-NLI and ANLI), outside the pipeline and with every variable under control.

B.3 A probe battery of minimal pairs

The battery consists of 121 premise-hypothesis pairs in four groups: eight *documented leakage cases*, distilled from pairs of the kind observed in the corpus, in which the premise describes a performance or an incidental fact and the hypothesis states an unrelated real-world event, so that the expected label is neutral; a set of *domain pairs* across sport, music, cinema, geography, science and history, built with the same recipe but with premises that genuinely describe their events; a set of *controls* (clear entailments, contradictions, unrelated pairs); and a series of *systematic manipulations* of one documented case, used in Section B.6. Each pair is scored once; E denotes the softmax probability of the entailment class.

=====	
SUMMARY	
=====	
Total test cases	: 121
Bias cases total	: 8
Strong leakage ($E > 0.5$)	: 5 (62.5%)
Suspicious ($E > 0.1$)	: 1 (12.5%)
Clean ($E \leq 0.1$)	: 2 (25.0%)
Control errors	: 2 / 7

Figure B.1: Summary of the probe battery: of the eight documented leakage cases, five score $E > 0.5$ and only two are clean.

Three headline results frame everything that follows (Figures B.1 to B.3). First, the leakage is real and strong: five of the eight documented cases score $E > 0.5$, up to $E = 0.973$ for a premise about a baseball team’s post-season record paired with the hypothesis that the MLB All-Star Game was held at SkyDome, $E = 0.949$ for a tenor’s performance paired with the location of the 2006 World Cup Final, and $E = 0.945$ for a guest performance paired with the location of the 2018 Champions League Final. Second, the model is not simply promiscuous: the domain pairs, spanning six topical areas, all score $E \approx 0.000$, and the entailment and contradiction controls are answered almost perfectly ($E \geq 0.99$ on the former). The failure is selective. Third, one control fails in a revealing way: the pair “The boiling point of water at sea level is 100 degrees Celsius” and “The Eiffel Tower was completed in 1889”, two well-formed encyclopedic facts with no connection whatsoever, is predicted *entailment* at $E = 0.560$. Whatever the model is checking, it is not logical consequence.

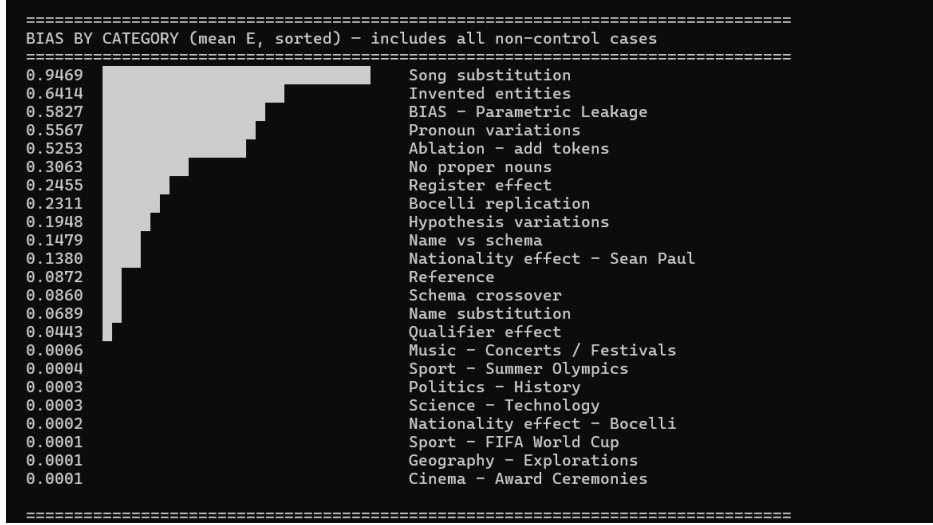


Figure B.2: Mean entailment probability by probe category. The categories that preserve the surface schema (song substitution, invented entities, pronoun variations) stay high; the genuine domain pairs at the bottom all score near zero.

CONTROL CASES				
[Control - Entailment	] pred=entailment	expected=entailment	✓	E=0.9968
[Control - Entailment	] pred=entailment	expected=entailment	✓	E=0.9948
[Control - Entailment	] pred=entailment	expected=entailment	✓	E=0.9912
[Control - Contradiction	] pred=contradiction	expected=contradiction	✓	E=0.0000
[Control - Contradiction	] pred=contradiction	expected=contradiction	✓	E=0.0000
[Control - Unrelated	] pred=entailment	expected=neutral	X WRONG	E=0.5601
[Control - Unrelated	] pred=contradiction	expected=neutral	X WRONG	E=0.0001

Figure B.3: Control cases. Entailment and contradiction controls are answered correctly with extreme confidence; one unrelated-facts control (boiling point of water against the Eiffel Tower) is wrongly predicted entailment at $E = 0.560$.

B.4 One case under the microscope

The rest of the appendix dissects a single documented case, chosen because half of the manipulation battery is built around it and because, as will become clear, it admits a naturally length-matched causal intervention. The pair is:

Premise. Jamaican rapper Sean Paul joined her as a special guest to perform their collaborative song, ‘No Lie’.

Hypothesis. The 2018 Champions League Final was held at the NSC Olimpiyski Stadium in Kyiv, Ukraine.

The premise describes a performance; the hypothesis states the location of a sporting event; nothing in the former textually entails the latter, and the expected label is neutral. The model predicts entailment at $E = 0.945$.

Two hypotheses compete to explain this. *Attestation bias* [11] holds that the model declares entailed any hypothesis stating a fact attested in its pre-training data, regardless of the premise; it is related to the hypothesis-only pathologies known for NLI datasets [15], and here it is prima facie plausible, since the hypothesis is a real, well-attested fact. *Surface schema matching* holds instead that the model has internalized frequent co-occurrence templates of pretraining text, of the form [ARTIST] + *perform* + [SONG] alongside a major-event statement, and ratifies the *form* of the co-occurrence without verifying the relation, a schema-level cousin of the shallow heuristics documented for NLI models [10]. The two hypotheses make different predictions under each instrument, which is what the case study exploits.

B.5 Attention: does the model consult its memory of the hypothesis?

If attestation bias were the mechanism, the premise would be a spectator: the prediction would be driven by the hypothesis tokens, and the attention reaching the classification token should be dominated by the hypothesis. The prediction is tested with attention flow [1], measuring the *H-dominance* of the pair: the share of the attention flow from [CLS] that lands on the hypothesis, relative to the flow landing on the whole input, with 0.65 as the empirical threshold above which a pair is considered hypothesis-driven.

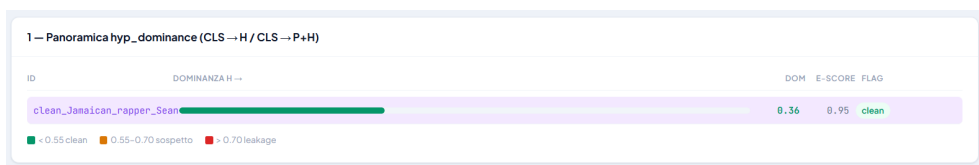


Figure B.4: H-dominance of the case pair: 0.36, well below the 0.55 clean band, despite the entailment score of 0.945.

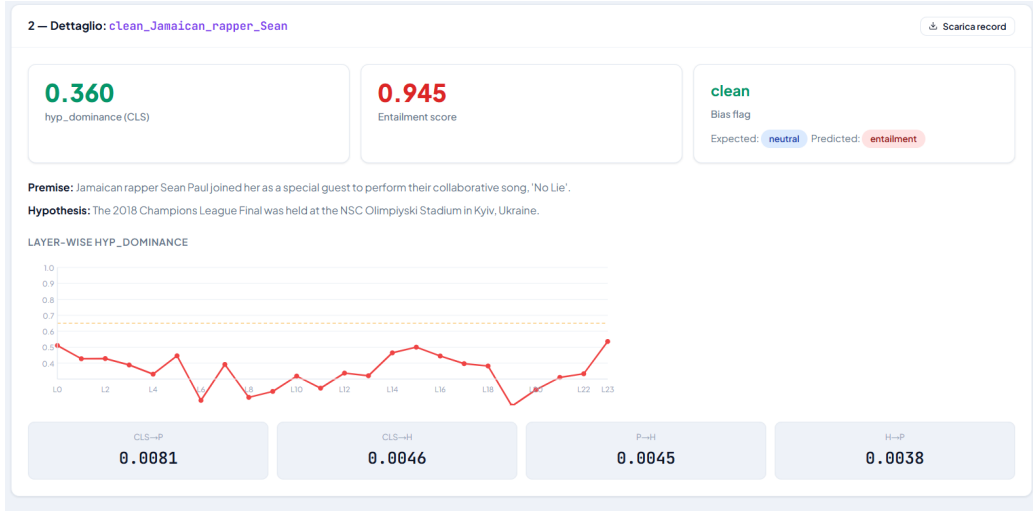


Figure B.5: Attention detail for the case pair: layer-wise H-dominance oscillates between 0.3 and 0.55 and never approaches the leakage band; the aggregate flows show [CLS] attending the premise (0.0081) nearly twice as much as the hypothesis (0.0046).

The result refutes the hypothesis on this case, consistently with the battery-wide finding. The pair’s H-dominance is 0.36, far below the 0.65 leakage threshold and below many clean control pairs; the layer-wise profile never crosses the band at any depth (Figure B.5); and the aggregate flows show the classification token attending the *premise* almost twice as much as the hypothesis (0.0081 against 0.0046), with the two cross-flows (premise to hypothesis and back) essentially symmetric. The model is not ignoring the premise and reciting a memorized fact; the premise participates actively in the error. Whatever the mechanism is, it reads both sides.

B.6 Integrated Gradients: the bias is structural

If the model were recalling the attested fact, feature attribution should concentrate on the entities that identify it: Kyiv, Ukraine, the stadium name. If it is matching a schema, attribution should concentrate on the tokens that make up the template on both sides, the performance frame in the premise and the event frame in the hypothesis, with the specific entities contributing little. Integrated Gradients [20] (50 steps, padding baseline, target = entailment logit) decides between the two.



Figure B.6: Token-level attribution for the case pair (green positive, red negative). The positive mass sits on the schema tokens of both sides; the location entities of the hypothesis (Kyiv, Ukraine) are negative.

TOP-10 TOKEN PIÙ INFLUENTI

#1	joined	↑ +2.2849	position 5
#2	Final	↑ +1.1799	position 27
#3	Stadium	↑ +0.9184	position 37
#4	2018	↑ +0.9124	position 24
#5	Ukraine	↓ -0.8864	position 41
#6	at	↓ -0.6758	position 30
#7	held	↑ +0.6722	position 29
#8	Jamaican	↑ +0.6572	position 1
#9	in	↓ -0.6131	position 38
#10	song	↑ +0.6071	position 15

Figure B.7: Top-10 tokens by attribution. The strongest contributor is the premise verb *joined* (+2.20); the hypothesis contributes through its event frame (*Final* +1.18, *Stadium* +0.92, *2018* +0.91, *held* +0.67), while *Ukraine* is the strongest negative (−0.89).

The verdict is unambiguous (Figures B.6 and B.7). The single strongest contributor to the entailment logit is the premise verb *joined* (+2.20), followed by the event-frame tokens of the hypothesis: *Final* (+1.18), *Stadium* (+0.92), *2018* (+0.91), *held* (+0.67); on the premise side, *Jamaican* (+0.66) and *song* (+0.61) complete the picture. The entities that would identify the attested fact point the other way: *Ukraine* is the strongest negative token in the input (−0.89), and *Kyiv* is negative as well. The model predicts entailment *despite* the location entities, not because of them, which is close to a direct refutation of factual recall on this pair, and exactly the signature of a template: a performance frame on one side, an event frame on the other, and the verdict issued on the strength of their co-occurrence. The layer-wise attribution map (Figure B.8) adds that this event-frame processing concen-

trates in the middle of the stack, with per-layer totals peaking around layers 9 and 14.

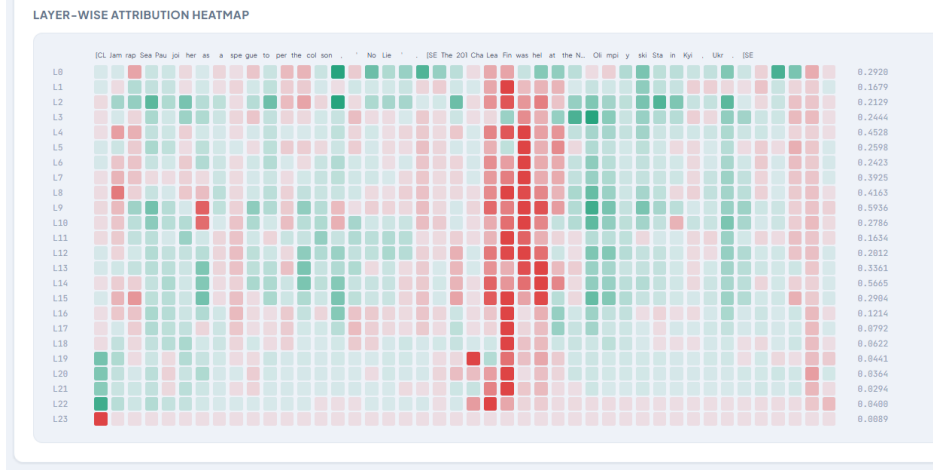


Figure B.8: Layer-wise attribution heatmap. The event-frame columns of the hypothesis (Final, was, held) dominate the middle layers, with per-layer attribution peaking around layers 9 and 14.

Behavioural corroboration. The manipulation series of the battery, all built on this case, agrees with the attribution picture on every axis. The score tracks schema completeness, not content: building the premise up token by token, the bare name scores $E = 0.081$ and the grammatical minimal variant 0.029, but the score jumps to 0.847 the moment the object phrase fills out to “perform collaborative song”, and the full sentence reaches 0.962. Content is interchangeable: replacing the song title changes nothing (four variants including the gibberish title ‘Xkzqw’ score between 0.934 and 0.956), pronouns matter little, and the schema fires with no shared entities at all: “A special guest performed a collaborative song at the ceremony.” entails the Kyiv hypothesis at $E = 0.876$. Even fully invented entities fire when the schema is intact (a fictional “Kryzbekian rapper Zorlan Vex” premise scores 0.961 against the real hypothesis), though not uniformly, a reminder that the phenomenon has irregular edges. Register matters in a way meaning does not: the caption-like “Sean paul perform song” fires at 0.705 while its grammatical twin, differing in a single capital letter, scores 0.029.

SCHEMA CROSSOVER - key diagnostic		
P (first 50)	H (first 50)	E
Sean Paul perform song	The 2018 Champions League Final was held at the NS	0.0291
Sean Paul perform song	The 2031 Florp Cup Final was held at the Zorbax Ar	0.0002
Sean Paul perform song	A sporting event took place at a stadium.	0.4846
Sean Paul perform song	Someone sang at a venue.	0.0010
Andrea Bocelli performed Nessun Dorma	The 2006 FIFA World Cup Final was played at the OL	0.0000
Andrea Bocelli performed Nessun Dorma	The 2031 Florp Cup Final was held at the Zorbax Ar	0.0010

Figure B.9: Schema crossover diagnostics: the minimal premise entails a vague event hypothesis with no shared entities ($E = 0.485$) more readily than the real one, and invented hypotheses behave irregularly.

The single most diagnostic behavioural result is the nationality sweep (Figure B.10). Prepending a nationality to the minimal premise generally suppresses the score: British, French, Brazilian and American variants all stay below 0.04. The one exception is *Ukrainian*, which restores it to 0.606. Ukraine is where the hypothesis’ final was held: the modifier that lexically bridges premise and hypothesis re-fires the inference that every other modifier kills. The model is demonstrably scoring surface co-occurrence between the two sides, which is also the natural reading of the water-Eiffel control failure, two sentences in flawless encyclopedic register with nothing else in common. The same sweep on the Bocelli case (Figure B.11) is flat at zero, confirming that the bridge needs a schema to amplify.

Nationality effect - Sean Paul - E by nationality modifier	
0.6064	Ukrainian Sean Paul perform song
0.1504	Italian Sean Paul perform song
0.0365	British Sean Paul perform song
0.0150	French Sean Paul perform song
0.0144	Brazilian Sean Paul perform song
0.0051	American Sean Paul perform song

Figure B.10: Nationality sweep on the minimal premise. Every modifier suppresses the score except *Ukrainian*, the one that lexically bridges to the hypothesis’ location.

Nationality effect - Bocelli - E by nationality modifier	
0.0004	British Andrea Bocelli performed Nessun Dorma
0.0004	German Andrea Bocelli performed Nessun Dorma
0.0003	Jamaican Andrea Bocelli performed Nessun Dorma
0.0001	French Andrea Bocelli performed Nessun Dorma
0.0000	Italian Andrea Bocelli performed Nessun Dorma

Figure B.11: The same sweep on the Bocelli variants is flat at zero: the lexical bridge amplifies a schema, it does not create one.

The schema hypothesis also retrodicts the corpus behaviour of Chapter 6 and Table B.1: the over-entailed corpus sentences (definitional statements,

theme sentences, measurement clauses, form text) are precisely low-content instances of encyclopedic schemas, the kind of premise whose form co-occurs with everything and whose content links to nothing.

B.7 Activation patching: where the signal lives

Attention flow and Integrated Gradients are correlational; the causal instrument is activation patching [12]: run the model on a clean pair and a bias-firing corrupted pair, replace the residual-stream activation of the clean run at a single (layer, position) with the corresponding activation of the corrupted run, and read how much of the score gap transfers. The method requires clean and corrupted inputs of identical token length, a constraint that frustrated a first attempt on the full-sentence pair (where a preliminary map nonetheless indicated the effect concentrating on premise positions in the early layers, with a second band around layers 9 to 12).

The register result of the battery hands the experiment a pair that satisfies the constraint perfectly: the grammatical minimal premise and its caption-register twin differ in a *single capital letter*, tokenize to identical lengths, and sit on opposite sides of the bias ($E = 0.029$ against $E = 0.705$, a gap of $+0.676$ produced by one character).

INPUT CLEAN (BASELINE CORRETTO)

PREMISE (CLEAN)

Sean Paul perform song

HYPOTHESIS (CLEAN)

The 2018 Champions League Final was held at the NSC Olimpiyski Stadium in Kyiv, Ukraine.

INPUT CORRUPT (BIASED)

PREMISE (CORRUPT)

Sean paul perform song

HYPOTHESIS (CORRUPT)

The 2018 Champions League Final was held at the NSC Olimpiyski Stadium in Kyiv, Ukraine.

Run Activation Patching

Figure B.12: The patching pair: clean and corrupted inputs differ in a single capital letter of the premise, with the hypothesis identical.

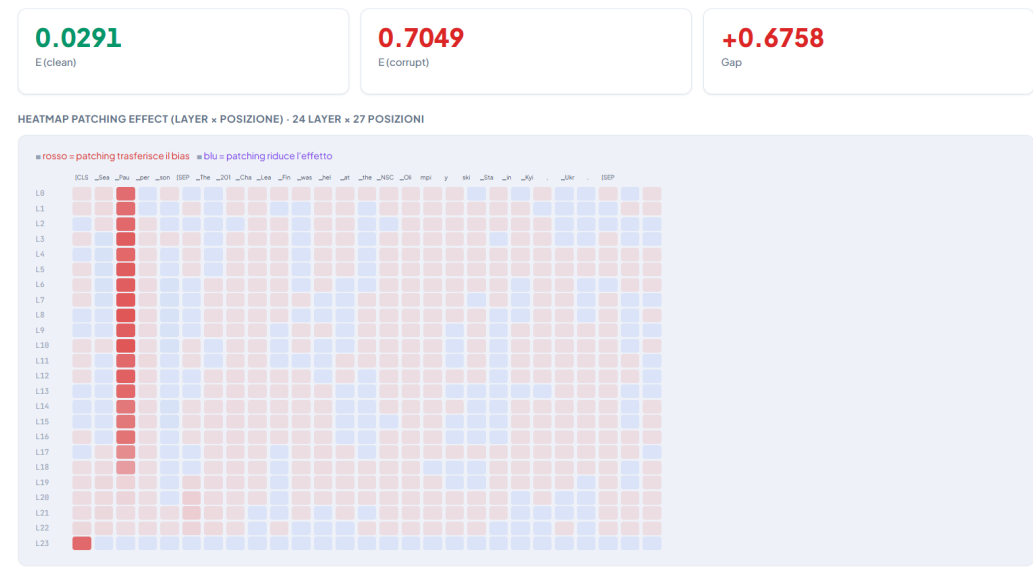


Figure B.13: Patching effect per (layer, position), 24 layers by 27 positions. The transferred effect is confined to the column of the altered token through layers 0 to 18, and appears at [CLS] only in the final layer.

The map (Figure B.13) is strikingly clean. The transferred effect forms a single vertical column at the position of the altered token, running from the first layer through roughly layer 18, and is essentially absent everywhere else until it appears at the [CLS] position in the final layer. Part of the column is expected by construction, since the two inputs differ at that position from the embedding onward; the informative content of the map is what does *not* happen. The signal does not spread: no other premise position, no hypothesis position, and no early [CLS] layer carries a transferable share of the bias. The register feature that flips the verdict is carried locally, in the residual stream of the token that bears it, through almost the entire depth of the network, and is integrated into the classification representation only at readout. A distributed, semantic recomputation of the premise would look different; a surface feature riding its own token until the final aggregation looks exactly like this. The causal evidence, as far as it currently reaches, points the same way as the attribution and the behaviour.

B.8 Provisional verdict and outlook

Two hypotheses entered the case study; one leaves refuted and one leaves standing at every checkpoint. Attestation bias, in the form “the model ignores the premise and ratifies attested hypotheses”, is contradicted twice

over: the attention flow shows the premise attended more than the hypothesis, and the attribution shows the fact-identifying entities pushing *against* the verdict. Surface schema matching is consistent with every instrument: the attribution mass sits on the template tokens of both sides, the score tracks schema completeness while contents are interchangeable, a single lexical bridge resurrects a killed inference, register outweighs meaning, and the causal map shows the decisive surface feature travelling locally in its token’s residual stream to a last-layer readout. What is still missing is the identification of the circuit itself: which attention heads read the schema, and where the two half-templates are matched against each other.

The implication for attribution systems does not wait for that step, because Chapter 6 already measured it. A scorer that ratifies surface schemas cannot be trusted with a per-item, absolute threshold on an open candidate set: any distractor sentence instantiating a frequent schema will clear it, which is exactly the noise collapse of Section 6.3. The BGE gate works because it removes schema-firing but content-irrelevant candidates before the cross-encoder scores them, a candidate-set fix for what this appendix indicates is an encoding-level bias.

Future work follows directly, with an applied goal: understanding *why* the model entails these pairs is the precondition for fixing the attributor at the source rather than around it. The next steps are per-head attention analysis and path patching on the length-matched pairs this battery generates, to move from the localized signal of Figure B.13 to an identified circuit; and, once identified, targeted fine-tuning on schema-breaking contrast pairs of exactly this kind, with the aim of recovering the precision of the gated configuration without the gate, improving the NLI attributor itself rather than its surroundings.



“What is there to be happy about? Job is not finished.”

— Kobe Bryant

